

Finance & Investment

CoF IIS 5

Winter Term 2025/2026

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Agenda

- 1. Fundamentals on the Corporate and on Corporate Finance**
2. Fundamentals on Financial Markets
3. Investment Valuation / Capital Budgeting
4. Calculation of Capital Needed
5. Corporate Financing
6. Corporate Analysis
7. International Corporate Finance
8. Derivative Financial Instruments

Learning Objectives Chapter 1

- Recapitulate fundamental corporate goals and Analyse implications for investment and financing
- Get to know different ownership structures and Analyse implications for investment and financing
- See and Understand differences in financial measurement of corporate activities
- Get to know main tasks of corporate finance sector
- Define important terms in corporate finance

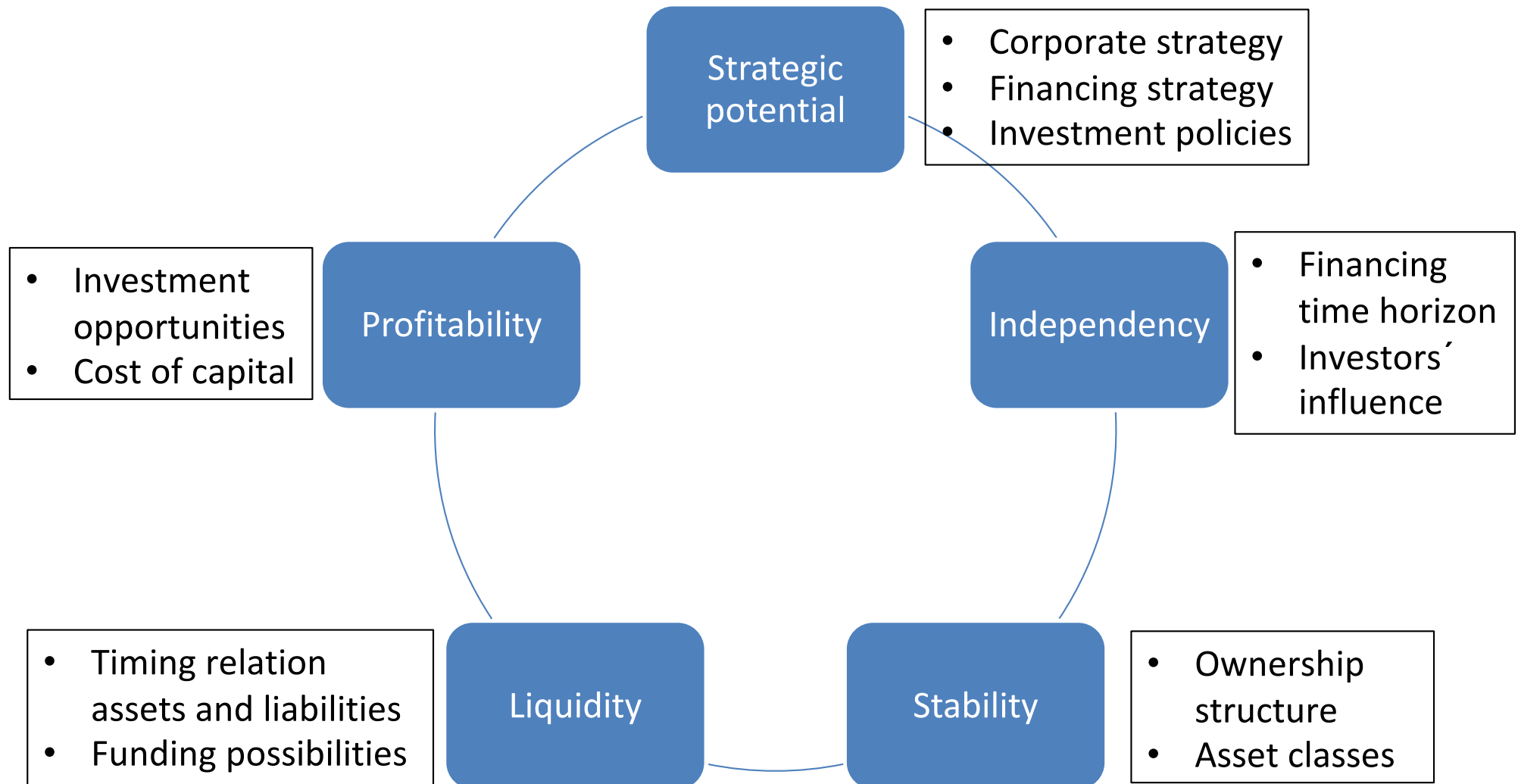
Content

1. Fundamentals on the Corporate and on Corporate Finance
 - 1.1 Corporate Goals and Ownership Structures
 - 1.2 Financial Aspects of Corporate Management
 - 1.3 Tasks of and Definitions in Corporate Finance

Corporate Goals

- General goals
 - Strategic potential
 - Financial goals
 - Profitability
 - Liquidity (short-term to medium-term)
 - Non-financial goals
 - Stability (medium-term to long-term)
 - Independency
- ? Discuss
- ? correlations between goals and
 - ? implications for decisions on financing and investing activities

Corporate Goals



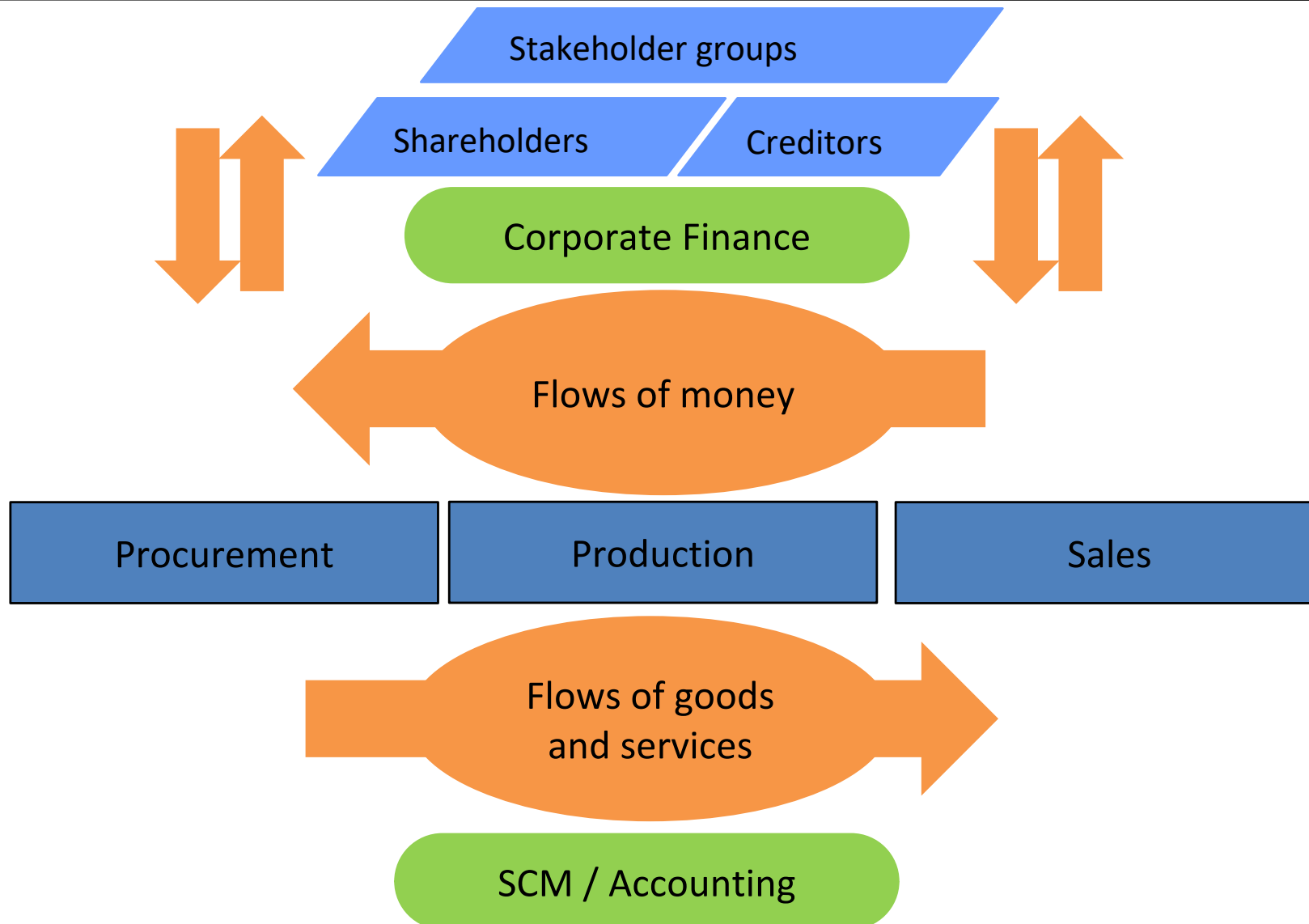
Ownership Structures

- Privately owned company
 - Partnership
 - Limited liability company
 - (Listed) Corporation
- ?
- What are the main characteristics and what are important differences of the different forms?
- ?
- What effects do these characteristics have on financing possibilities and on investing activities?

Ownership Structures

	Privately owned company	Partnership	Limited liability company	(Listed) Corporation
Corporate activity of owner	Yes	Partially	Partially	Typically not
Distinction owner / manager	No	No	No	Yes
Personal liability owner	Yes	Yes	No (partially in case of personal guarantees)	No
Financing via capital markets	No	No	Partially	Yes
Volume of capital sources	Limited	Limited	Partially unlimited	Unlimited
Investment opportunities	?	?	?	?

Fundamental Business Model



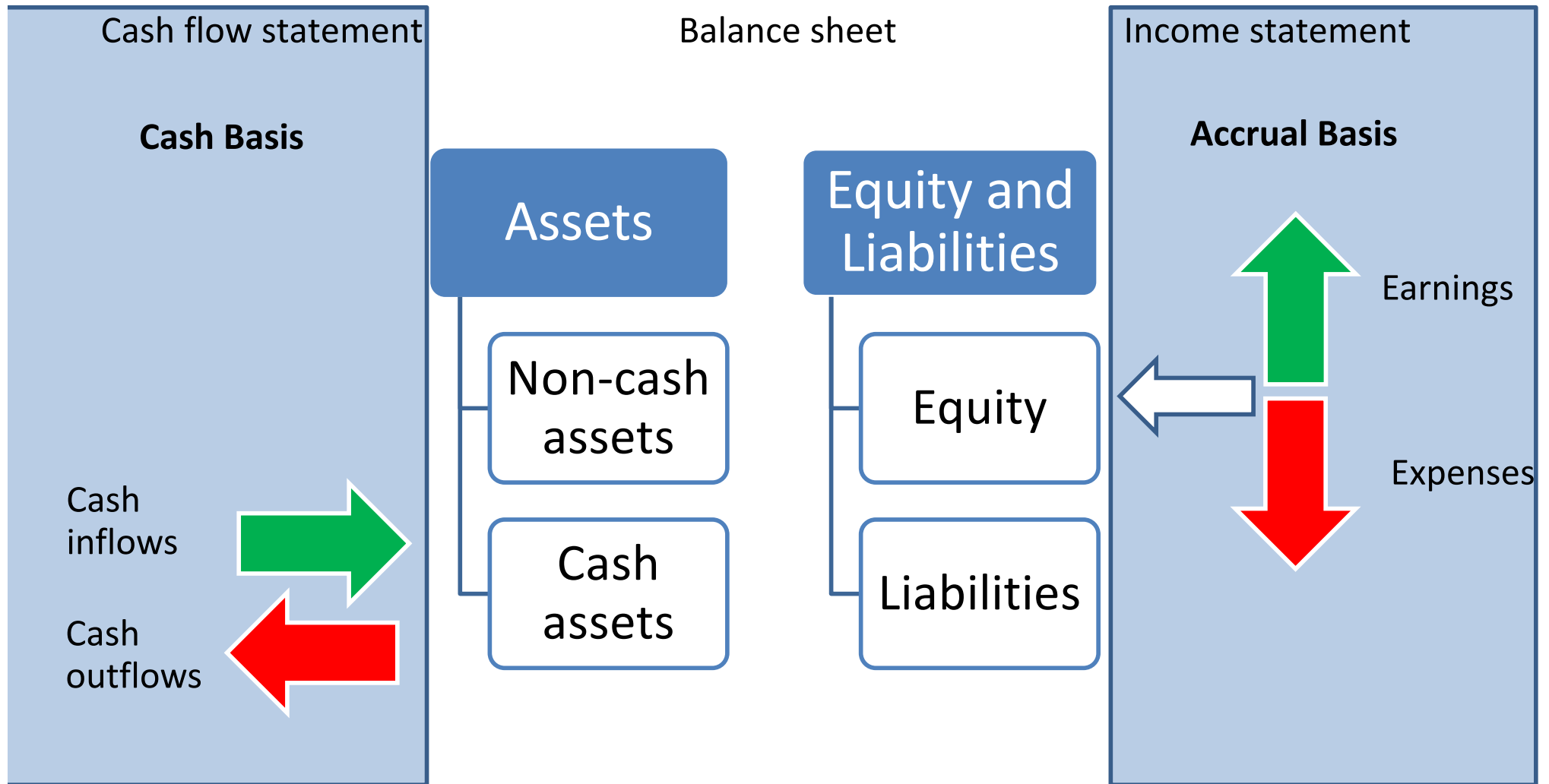
Financial Statements Elements

- Balance sheet
- Income statement
- Statement of changes in equity
- Cash flow statement
- Segment reporting
- Notes
- Management discussion and analysis
- Corporate Governance declaration

Measurement of Balances

- There are different ways to measure the existing balances of a company at a certain point in time:
 - The accounting view (bookkeeping approach) is based on accrued cash flows (cash inflows and cash outflows) that have occurred in the past or that will occur in future. These cash flows are stored in the balance sheet.
 - The financing view (monetary approach) is not accruing positive or negative cash flows or stocks but taking them directly into account.

Measurement of Changes in Stock



Measurement of Changes in Stock

- There are different ways to measure the changes in stock of a company in a given period:
 - The accounting view (bookkeeping approach) is documenting changes in equity with earnings and expenses, which are based on accrued cash flows (“accrual principle of accounting”).
 - The financing view (monetary approach) is taking positive or negative cash flows (cash inflows and cash outflows) into account.
- Both views typically lead to different results, only in perpetuity the overall results will be identical.

Corporate Finance Tasks

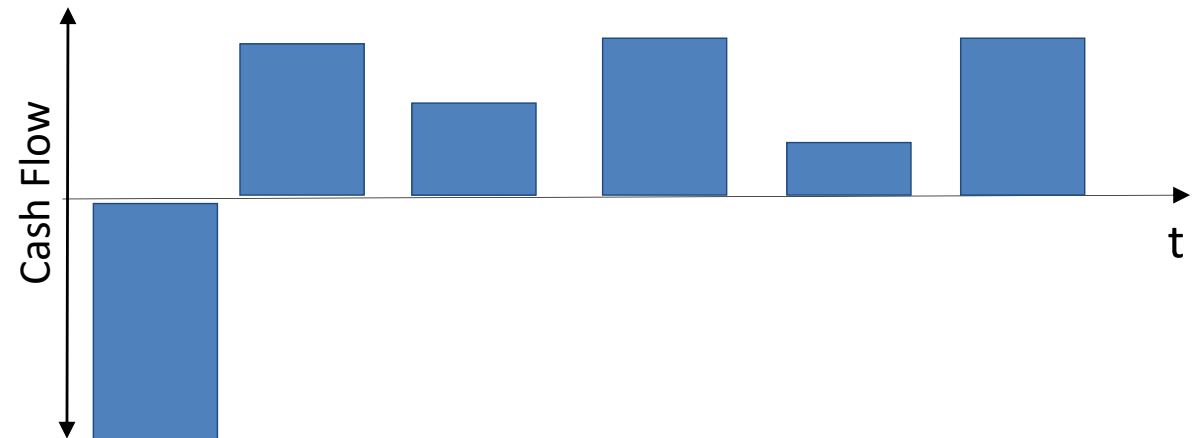
- Guarantee of liquidity (= ability to meet all financial obligations at any time)
- Provide operative business units and processes with the required amount of liquidity (= capital needed)
- Install and maintain relationships to financing institutions
- Optimisation of finance costs and risks
- Generation of positive contributions through financial activities
- Reduction of currency risks from the operative business and from financing activities

Corporate Finance Process

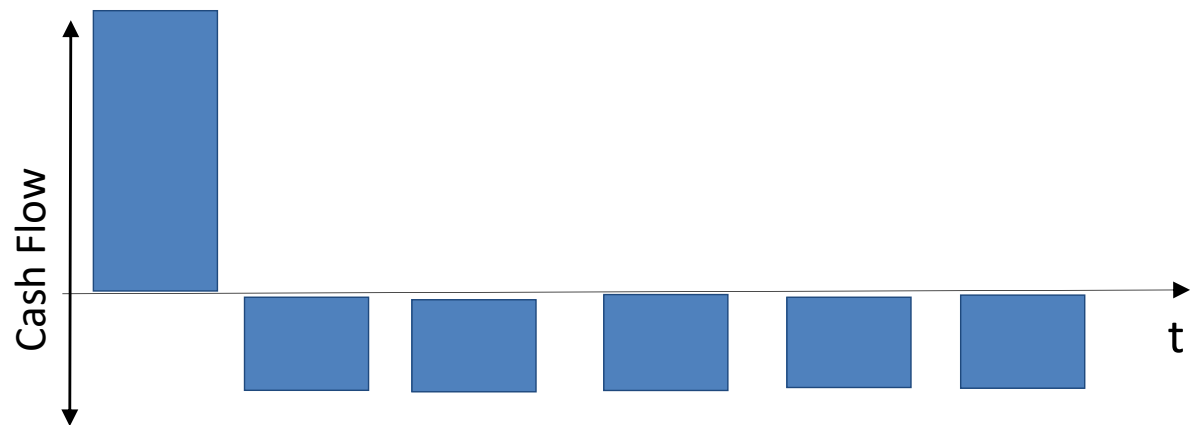
1. Identification of needed investments
2. Calculation of investment volume
3. Calculation of capital needed
4. Identification of financing possibilities
5. If capital needed is covered => step 7; if not => step 6
6. Reduction of capital needed possible?
If not: Need for measures to increase available capital
7. Planning of Cash flows (Cash inflows and Cash outflows)
8. Settlement of Cash flows

Investment & Finance

Investment: series of cash flows starting with a cash outflow



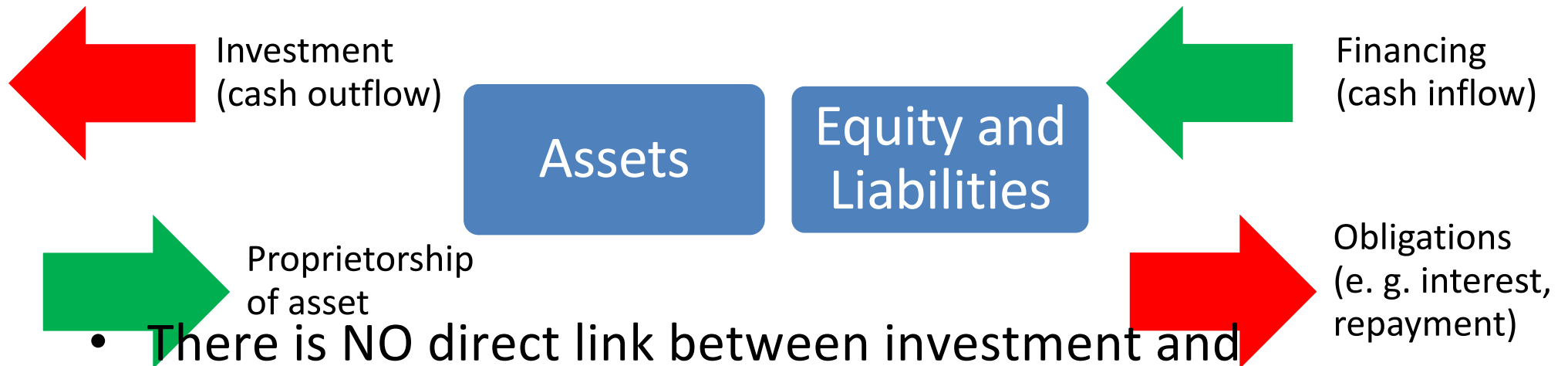
Finance: series of cash flows starting with a cash inflow



Terms and Definitions

- An investment is the acquisition of an asset for future earnings and profits. It is represented by a cash stream that starts with a cash outflow.
- Financing provides funds to a corporate. It is represented by a cash stream that starts with a cash inflow.
- Corporate finance comprises both aspects and is the overall generic term. (Note: There are other definitions)
- A disinvestment means the disposal of an asset.
- The repayment of a financing fund can be called refunding.

Let's combine Investment and Financing?



- There is NO direct link between investment and financing. Decisions on both aspects are separated.
- The opposite terms are disinvestment and refunding, but they cannot either be linked adequately (e. g. depreciation reduces asset value but is not disinvestment and equity funds are not to be repaid).

Learning Objectives Chapter 1

- ✓ Recapitulate fundamental corporate goals and
Analyse implications for investment and financing
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measurement of corporate activities
- ✓ Get to know main tasks of corporate finance sector
- ✓ Define important terms in corporate finance

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Learning Objectives Chapter 2

- Get to know relevant definitions in the financial market context
- Learn and Structure different products in financial markets
- Understand theoretical derived functions and assumptions of financial markets and Discuss applicability in practise
- Hear about legislation for and supervision of financial markets
- Develop a basic model of a financial market and Derive consequences for corporate decisions

Content

2. Fundamentals on Financial Markets

2.1 Financial Market Terms and Definitions

2.2 Functions and Assumptions of Financial Markets

2.3 Legislation and Regulation in Financial Markets

2.4 Basic Model of a Financial Market

Financial Market – Terms and Definitions

- Financial Markets combine supply and demand for financial products, also called as financial securities.
- Main forms are:
 - Money market offers short-term investment and financing (up to 12 months).
 - Capital market (in a wider sense) is a market for medium-term and long-term equity and debt.
 - Capital market (in a narrower sense) is a market for financial securities (shares and bonds).
 - Capital market (in the narrowest sense) is an organised security market (exchange of financial products).

Financial Market – Terms and Definitions

- Other terms that are frequently used are:
 - Credit market (in a wider sense) is a market for debt capital.
 - Credit market (in a narrower sense) is a market for non-securitised debt capital.
 - The market for financial derivatives trades assets like options, futures, swaps and structured products.
- The use and concrete meaning in praxis are not always clear nor consistent.

Financial Market – Participants

- “Regular” companies
- Financial institutions
 - Insurance companies
 - Pension funds
 - Capital investment companies
 - Banks
 - Leasing companies
- Private persons
- National and international legislation and supervision authorities
- ...

Financial Market – Products

- Direct products
 - Equity
 - Debt finance
 - Corporate bonds
 - Convertible bonds
- Derivative products
 - Forwards and futures
 - Options
 - Swaps
- Information and Intermediation products
 - Equity issues
 - Debt arrangements
 - Derivative development („financial chemistry“)

Financial Market – Functions

- Volume transformation: Transfer of liquidity between economic entities
 - The single/individual investment into a company is relative small.
 - The accumulated amount of invested money is relative big.
- Time transformation: Transfer of liquidity over time
 - The investment into a company can be time-limited.
 - The liquidity insertion (cash) itself can be contributed time-unlimited.
- Risk transformation
 - The investment into one company is relative risky.
 - But a spread of risks is possible through investment alternatives and diversification.
- Information transfer
 - Production of information supply
 - Satisfaction of information needs

Financial Market – Assumptions

- Unlimited funding and investing
 - Debtors can obtain any desired kind and volume of finance.
 - Investors can invest funds unlimitedly.
- Unique interest rate
 - The interest rate for investing and for borrowing capital is identical.
 - In the interest rate all relevant and available information is included.
- No transaction cost
 - Every acquisition and sale of financial securities is at 0 cost.
 - This is also valid for lending financial securities.
- Atomistic structure
 - A finite, but large enough number of participants is active.
 - Single actions of the market participants do not change market prices.
- Information efficiency
 - All market participants have the same information.
 - All relevant information is included in the prices for financial securities.

Arbitrage in financial markets

- Arbitrage describes the simultaneous buying and selling of goods/securities in different markets to benefit from pricing differences.
- An arbitrage opportunity has a Net Present Value > 0 .
- As long as arbitrage opportunities exist, investors will use them, until the price differences (=arbitrage opportunity) disappear.
- Therefore in “normal” (sometimes called “information efficient”) financial markets arbitrage cannot exist.

Arbitrage example

- A (zero-coupon, i. e. without ongoing payments) bond with a face value of 1,000 € and a duration of 1 year has a market price of 940 €. The interest rate $i = 5\%$.
- ? What is the “right price” for the bond?
- ? How would you act in this financial market?
- ? What will happen to the market price of the bond?
- The price for the bond will increase to its intrinsic value of 952.38 €. All investors will pay the same price, arbitrage opportunities are eliminated.

Arbitrage example (modified)

- A (zero-coupon, i. e. without ongoing payments) bond with a face value of 1,000 € and a duration of 1 year has a market price of 960 €. The interest rate $i = 5\%$.
- ? What is the “right price” for the bond?
- ? How would you act in this financial market?
- ? What will happen to the market price of the bond?
- The price for the bond will decrease to its intrinsic value of 952.38 €. All investors will pay the same price, arbitrage opportunities are eliminated.

The Law of One Price

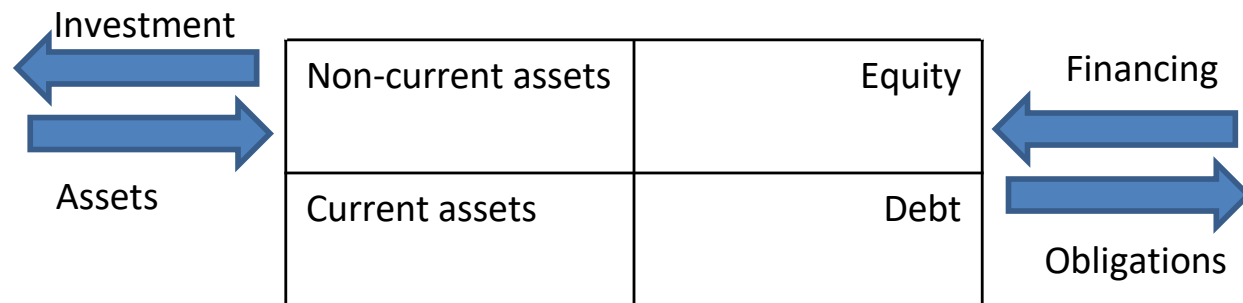
- As financial markets are arbitrage-free, for financial securities only one price can exist.
- The law of one price demands that equivalent financial securities are traded for the same price in different markets.
- Therefore this one price is the reference value for taking NPV based financial decisions.

Separation of Investment and Financing

- As no arbitrage opportunities can exist, financial actions and/or security transactions in “normal” markets do not generate value.
- Therefore financing decisions can be analysed separately from investing decisions.
- In other words, investment decisions should only respect the NPV rule regardless the intended or the effectively executed financing.

Separation of Investment and Financing

- There is **no** relation between investment and financing
- Decisions on investment and financing can therefore be separated.



Separation Example

1. A company can invest 10 Mio. € into a project. After one year the return is 12 Mio. €.

? What is the NPV @ $i = 10\%$? What is your decision?

➤ $-10 + [12 / (1.1)^1] = 0.91$

2. The company can get financing of 5 Mio. € for one year @ $i = 10\%$.

? How is the cash flow structure changed? How is the decision affected?

➤ $[-10 + 5] + [(12 - 5 * (1.1)^1) / (1.1)^1] = -5 + 5.91 = 0.91$

➤ The cash flow structure is changed.

➤ But: The NPV and therefore the original decision (execution of the project) are not changed.

Legislation and Regulation

- [Federal](#) Ministry of Finance
- [BaFin](#): Federal Financial Supervisory Authority
 - Supervision of banks, insurances and publicly listed companies
 - Supervision of open end funds
 - Supervision of prospectus of closed end funds
- National laws
 - Kreditwesengesetz
 - Kapitalanlagegesetzbuch
 - Wertpapierhandelsgesetz
 - Geldwäschegesetz
- International influence
 - Basel III
 - Financial transaction tax
 - Public supervision of banking under ECB control
 - Derivative regulation
 - Minimum risk requirement for banks

Financial Market Basic Model

- Without the existence of a financial market, corporate decisions for individual investors cannot be brought in line.
- Financial markets enable companies to decide without respecting the owners' individual preferences.
- The benefit of a financial market can be shown in a two step model with existing investment projects and where decisions are based on utility functions
 1. without financial market (i. e. only investment market and consumption market)
 2. additionally with a financial market

Model description (with two markets)

- The model consists of two periods t_0 and t_1 .
 - The initial budget B_0 in t_0 is 100.
 - An individual can either consume in t_0 (“ c_0 ”) or invest (parts of) the budget for higher consumption in t_1 (“ c_1 ”).
 - The following asset investments A, B, C, D can be done:
- ? Calculate c_0 , c_1 and c_0+c_1 for the investment alternatives and illustrate the results in a c_0 - c_1 -coordinate system.

Project	Invest. I_0	Return I_1	c_0	c_1	Σc_0+c_1
A	- 0	+ 0	100	0	100
B	- 40	+ 60	60	60	120
C	- 80	+ 108	20	108	128
D	- 100	+ 124	0	124	124

Decision taking with utility functions

- Investor 1 has the utility function $U_{In1} = c_0^{0.5} * c_1^{0.4}$
- ? Discuss the appropriateness of the utility function.
- Investor 2 has the utility function $U_{In2} = c_0^{0.1} * c_1^{0.8}$
- ? Discuss differences between U_{In1} and U_{In2} .
- ? Illustrate the utility functions in the coordinate system.
- ? Calculate U_{In1} and U_{In2} for the investment alternatives.

Project	Invest. I_0	Return I_1	c_0	c_1	$\Sigma c_0 + c_1$	U_{In1}	U_{In2}
A	- 0	+ 0	100	0	100	0	0
B	- 40	+ 60	60	60	120	39.84	39.84
C	- 80	+ 108	20	108	128	29.10	57.13
D	- 100	+ 124	0	124	124	0	0

Introduction of a financial market

- Consumption c_0 can now be substituted by savings @ 10% or increased by financing with debt @ 10%.
- ? Include capital market lines in the coordinate system, derive the capital market function and interpret movements up and down the line.
- $c_1 = f(c_0) = 110 - 1,1 * c_0$
- ? Which decision is now optimal for investors 1 and 2?
- The first decision is no longer on consumption or investment but on maximising the initial budget without respecting individual preferences.
- The decision on consumption (as well as savings/debt) is done in the second step based on individual preferences.

Calculation of budget, plans and utilities

- ? Calculate the new budget $B_0 (=I_{1 \rightarrow 0} + c_0)$ for the projects.
- With $U = c_0^\alpha * c_1^\beta$ individual consumption is calculated by:

$$c_0^* = \frac{\alpha}{\alpha + \beta} * B_0 \quad c_1^* = (1 + i) * \frac{\beta}{\alpha + \beta} * B_0$$

- ? Develop and analyse the individual plans (consumption, savings/debt) for investor 1 and 2 only for project C.
- ? Calculate U_{ln1}^* and U_{ln2}^* . Compare them with U_{ln1} and U_{ln2} .

Project	I_0	$\Rightarrow I_1$	$I_{1 \rightarrow 0}$	B_0	$c_0^*_{ln1}$	$c_1^*_{ln1}$	U^*_{ln1}	$c_0^*_{ln2}$	$c_1^*_{ln2}$	U^*_{ln2}
A	0	0	0	100						
B	40	60	54.54	114.54						
C	80	108	98.18	118.18	65.66	57.78	41.05	13.13	115.55	57.82
D	100	124	112.72	112.72						

Summary – The Fisher Separation

- Financial markets enable companies to decide without respecting the owners' individual preferences.
- The investment decision is not depending on the individual consumption preference and can therefore be separated and executed by a company or by the management without additional personal information.
- In case of perfect financial markets companies “only” need to maximise the initial budget by choosing the optimal investment project(s).
- The successive decision on consumption is depending on individual preferences. Based on that, individuals will either increase their future consumption by saving money or increase their present consumption by taking debt.

Learning Objectives Chapter 2

- ✓ Get to know relevant definitions in the financial market context
- ✓ Learn and Structure different products in financial markets
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Learning Objectives Chapter 3

- Recapitulate basic formulas for financial tasks
- Understand the importance of the time value of money
- Understand purposes, types process and tasks of investments
- Get to know static methods of capital budgeting, Apply them to different situations and Assess them
- Get to know dynamic methods of capital budgeting, Apply them to different situations and Assess them
- Calculate lifetime of investments
- See ways to respect risk in investment decisions

Content

- 3. Investment Valuation / Capital Budgeting
 - 3.1 Mathematical Fundamentals of Capital Budgeting
 - 3.2 Purposes and Types of Investments
 - 3.3 Investment Process
 - 3.4 Tasks of Capital Budgeting
 - 3.5 Static Methods
 - 3.6 Dynamic Methods
 - 3.7 Decisions on the Ordinary Lifetime of Investments
 - 3.8 Capital Budgeting under Uncertainty

Time Value of Money

- The value of money is changing over time.
- In other words: The value of one unit in currency today is not identical to its future value.
- This is due to interest that can be earned by saving money in a financial market or that has to be paid for debt obligations.
- This effect is named “The time value of money”.
- It is represented in discounting (to get the present value) or compounding (to get the future value) payments.
- The time value of money is determined by interest rate and length of the (discounting or compounding) period.
- The interest rate is depending on market and company-individual factors such as ownership, risk and duration.

Present Value (PV)

- The calculation of the present value (PV) is a mathematical technique to place a current value to future cash flows as of today.
- It is computed by discounting future cash flows (CF) at an interest rate reflecting the company's cost of capital (i).

$$PV = \frac{CF_n}{(1 + i)^n}$$

with

PV: Present Value

CF_n : Cash flow of period n

i: Discount rate

n: Number of periods

Future Value (FV)

- The calculation of the future value (FV) is a mathematical technique to place a value to current cash flows as of a point in time in future.
- It is computed by compounding current cash flows (CF) at an interest rate reflecting the company's cost of capital.

$$FV = (1 + i)^n * CF_0$$

with

FV: Future Value

CF_0 : Cash flow of period 0

i: Compound rate

n: Number of periods

Net Present Value (NPV)

- The net present value (NPV) is the total sum of all future cash flows discounted to the PV, reduced for the initial (as well as potentially additional) investment(s).

$$\text{NPV} = -\text{CF}_0 + \sum_{t=1}^n \frac{\text{CF}_t}{(1+i)^t}$$

with

NPV: Net Present Value

CF_0 : Initial investment

CF_t : Cash flow of period t (which is potentially negative)

i : Discount rate

n : number of periods

Examples

- ? A company buys a machine today for 100 and receives the following payments: $Y_1 = 30$, $Y_2 = 40$, $Y_3 = 50$. Calculate the NPV @ $i = 10\%$.

	10%	1	2	3
	-100	30	40	50
-2,10	-100	27,27	33,06	37,57

- ? A company buys a machine today for 100 and receives the following payments: $Y_1 = 50$, $Y_2 = 40$, $Y_3 = 30$. Calculate the NPV @ $i = 10\%$.

	10%	1	2	3
	-100	50	40	30
1,05	-100	45,45	33,06	22,54

Annuities (1)

- The transformation of several (not infinite) identical payments into one present value (PV) payment can be calculated by using the annuity value factor.
- Thus, to aggregate constant cash flows over a distinct period of time to the PV, the formula can be shortened:

$$PV = CF * \frac{(1 + i)^n - 1}{(1 + i)^n * i}$$

with

PV: Present Value

CF: (identical) Cash flow of periods 1 to n

i: Discount rate

n: Number of periods

Annuities (2)

- The transformation of one present value payment into several (not infinite) identical future payments can be calculated by using the recovery rate.
- Thus, to distribute one payment into several future payments the PV formula can be modified:

$$CF = PV * \frac{(1 + i)^n * i}{(1 + i)^n - 1}$$

with

PV: Present Value

CF: (identical) Cash flow of periods 1 to n

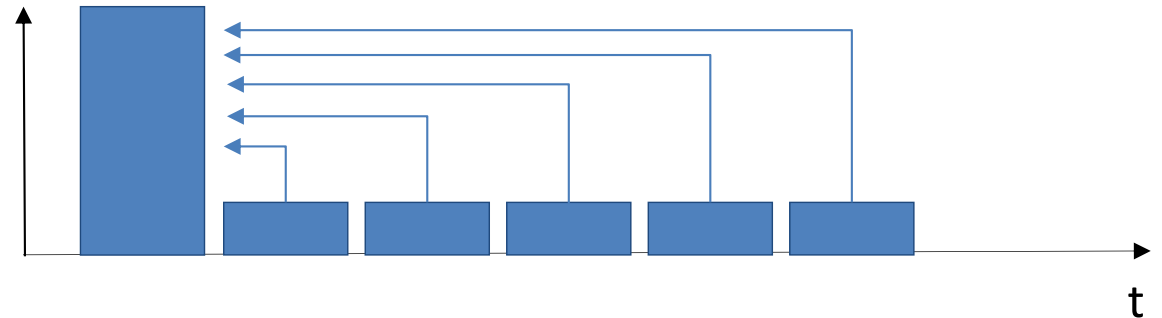
i: Discount rate

n: Number of periods

Annuities (1 and 2)

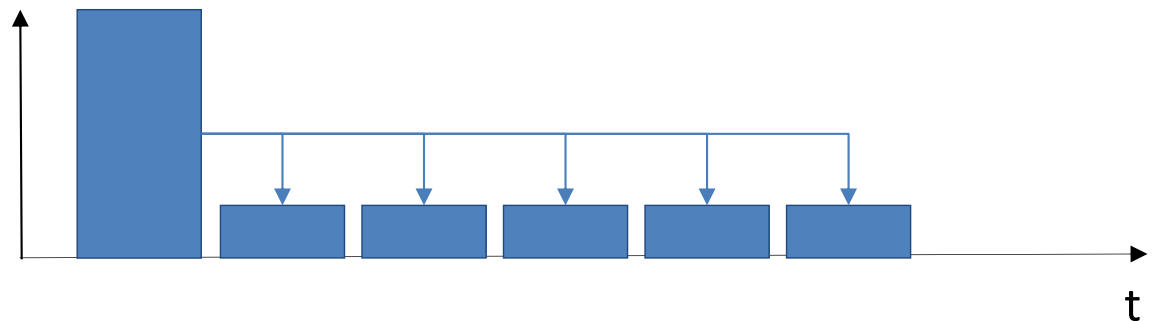
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$$PV = CF * \frac{(1 + i)^n - 1}{(1 + i)^n * i}$$



2

$$CF = PV * \frac{(1 + i)^n * i}{(1 + i)^n - 1}$$



Examples

- ? An annual rent of 5,000 € with a duration of 10 years shall be replaced. What is the Present Value with an interest rate of 5%?

$$PV = CF * \frac{(1 + i)^n - 1}{(1 + i)^n * i} = 5,000 * \frac{(1 + 0.05)^{10} - 1}{(1 + 0.05)^{10} * 0.05} = 38,609$$

- ? A loan of 100,000 € shall be paid back completely in 10 years with an interest rate of 5%. Calculate the annual payment.

$$CF = PV * \frac{(1 + i)^n * i}{(1 + i)^n - 1} = 100,000 * \frac{(1 + 0.05)^{10} * 0.05}{(1 + 0.05)^{10} - 1} = 12,950$$

Perpetuities

- In case of constant, infinite cash flows the PV formula can be shortened as follows:

$$PV = \frac{CF}{i}$$

with

PV: Present Value

CF: (identical, infinite) Cash flow of periods 1 to ∞

i: Discount rate

Relation Result, Return and Value

Based on the idea of perpetuity there are three different views on the valuation of a cash flow / an investment:

1. What is the Result?: Given the investment volume (value) and the return (profitability), the result is:

$$\mathbf{Result\ (profit\ or\ cash\ flow) = Investment\ (or\ Value) * Return\ i}$$

2. What is the Return?: Given the investment volume (value) and the result, the return (profitability) is:

$$\mathbf{Return\ i = \frac{Result\ (profit\ or\ cash\ flow)}{Investment\ (or\ Value)}}$$

3a. What is the value?: Given the result and the return (profitability), the value is:

$$\mathbf{Value = \frac{Result\ (profit\ or\ cash\ flow)}{1} * \frac{1}{Return\ i}}$$

3b. Instead of the relative return (profitability) i, frequently a multiplier m is used:

$$\mathbf{Value = Result\ (profit\ or\ cash\ flow) * Multiplier\ m}$$

Examples

1. A bond has a face value of 100,000 € and returns @ $i = 4\%$.
What is the annual interest in €?
2. A bond has a face value of 100,000 € and infinite annual payments of 4,000 €. What is the implicit discount rate?
3. a) A bond has an annual payment of 4,000 €.

What is the value of the bond with a discount rate of 4%?

 b) A bond has an annual payment of 4,000 €.

What is the value of the bond with a multiplier of 25?

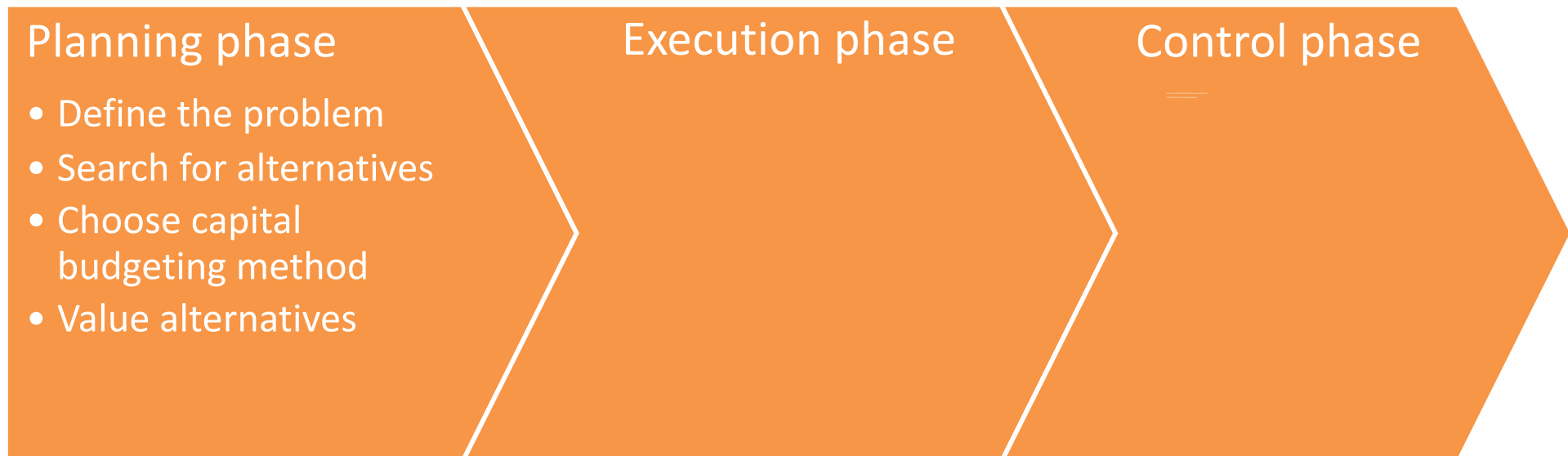
Purposes of Investments

- Economic purposes of investments
 - Creation of fundament for future growth (potential)
 - Access to new markets and clients
 - Development of new products
 - Improvement of efficiency of business processes
- Financial purposes of investments
 - Increase of operating cash flow through investments
 - Optimisation of corporate situation with respect to strategic corporate finance
 - Creating basis for organic growth and/or M&A activities

Types of Investments

- An investment is a cash stream that starts with a cash outflow.
- Types of investments (according to timeline)
 - Initial investments
 - Replacement investments
 - Enlargement investments
 - Rationalisation investments
- Types of investments (according to asset class)
 - Investments in tangible assets
 - Investments in intangible assets
 - Investments in financial assets / securities

Investment Process



Investment Valuation Criteria

- The following criteria could be used to value investment alternatives:
 - Maximising the value of a cash stream
 - Spending money (altruism)
 - Gaining image
 - Fulfilling personal intentions
- The focus of this course is on maximising the value.

Capital Budgeting

- Capital budgeting, also called investment valuation or investment appraisal, is an internal calculation.
- It helps to answer three different questions:
 1. An execution decision refers to the absolute advantage:
Shall one investment be executed or not?
 2. A selection decision refers to the relative advantage:
Which (out of two different and mutually exclusive) investment should be executed?
➤ The relative advantage is only analysed if the investment per se is advantageous.
 3. Capital budgeting can also be used to determine the right point in time to replace an existing investment (asset).

Static Methods – General Description

- Static methods can be used to calculate the absolute advantage of an investment as well as the relative advantage of (two or more) different investment projects.
- The calculation of the investment projects is based on accounting figures (costs and results / earnings and expenses) rather than on financial figures (cash flows).
- Static methods consider only one period, i. e. they combine the investment project to its “average” figures and do not respect differences in single periods.
- Static methods therefore do not consider the time value of money.
- Nevertheless they are widely used as they are easy to apply (and to understand).

Cost Comparison Method – Description

- The cost comparison method compares the average cost of two different investment alternatives.
- The alternative with lower average costs is preferred.
- Only the costs (and earnings) that are directly generated by the investment decision should be taken into account.
- Cost comparison is only reasonable if time horizon and average earnings are identical.
- It might be necessary to adjust the alternatives to different production capacities or to calculate the average cost per piece.
- Additionally the identity of earnings must not only be in volume but also in quality.

Cost Comparison Method – Description

- Ongoing costs p. a. (which are accounted to the average period automatically) are e. g.
 - Material cost
 - Personal cost
 - Maintenance cost
 - ...
- The Initial investment represents “one-time costs” that need to be transformed to the average period:
 - Depreciation
 - Cost of capital

Cost Comparison Method – Description

- The initial investment of $t = 0$ is transformed into Depreciation cost and Cost of capital that are taken into account in the average period.
- Depreciation cost is calculated with the straight-line method:

$$D = \frac{I_0 - L}{n}$$

- Cost of capital is typically calculated with an interest rate on the invested capital (capital employed) based on the average method:

$$\text{Cost of capital} = i * CE = i * \left(\frac{I_0 - L}{2} + L \right) = i * \frac{I_0 + L}{2}$$

with

D:	Depreciation
n:	Number of periods
I_0 :	Investment
L:	Liquidation proceeds
CE:	Capital invested (Capital employed)
i:	Interest rate

Depreciation and Cost of Capital – Example

- A company acquires a machine for 100,000 €, the expected ordinary lifetime is 10 years, liquidation proceeds are 20,000 €. The interest rate for the cost of capital is 5%.
- ? Calculate the annual depreciation cost and the annual cost of capital.

$$\text{➤ } D = \frac{100,000 \text{ €} - 20,000 \text{ €}}{10 \text{ years}} = 8,000 \frac{\text{€}}{\text{year}}$$

$$\text{➤ } \text{Cost of Capital} = 5\% * \frac{100,000 \text{ €} + 20,000 \text{ €}}{2} = 3,000 \frac{\text{€}}{\text{year}}$$

Cost Comparison Method – Example

- A company needs to decide between two new sites A and B.
 - The produced volume is 8,000 units.
 - The interest rate for cost of capital is $i = 10\%$.
 - The final product can be bought externally for 90 €.
- ? Which decision is the right one?

Cost Comparison Method – Example

		Site A	Site B
Capital employed			
	Investment	300,000 €	480,000 €
	Economic lifetime	6 years	6 years
	Liquidation proceeds	0 €	60,000 €
	Capacity	6,000 units	8,000 units
Fixed costs p. a.			
	Maintenance cost	40,000 €	50,000 €
	Other fixed costs	3,000 €	5,000 €
Variable costs p. a.			
	Material costs	400,000 €	450,000 €
	Personal costs	20,000 €	80,000 €
	Other variable costs	30,000 €	40,000 €

Cost Comparison Method – Solution

		Site A	Site B
Capital employed		300,000 € / 2 = 150,000 €	(480,000 € + 60,000 €) / 2 = 270,000 €
	Cost of capital @ i = 10%	15,000 €	27,000 €
	Depreciation cost	300,000 € / 6 = 50,000 €	(480,000 € - 60,000 €) / 6 = 70,000 €
	Fixed costs	43,000 €	55,000 €
	Variable costs	450,000 €	570,000 €
	Additional costs (capacity adjustment)	2,000 units * 90 €/unit = 180,000 €	n/a
➤ Based on this calculation the investment in site B is advantageous!	Total costs	738,000 €	722,000 €

Cost Comparison Method – Assessment

- Cost comparison is only reasonable if time horizon and average earnings are identical.
- The cost comparison method does not respect timing differences.
- The preferableness of an investment can depend on the number of produced units (which is easy to manipulate).
- The differentiation of fixed and variable costs is in praxis difficult.
- The market / sales side is neglected.

Profit Comparison Method – Description

- The profit comparison method calculates the average profit of investment alternatives.
- The profit comparison method is also only applicable if projects have the same time horizon, as it makes a clear difference whether an average profit is earned once or more often.
- An adjustment in capacity might be necessary.
- The decision rules for investment projects are:
 - An investment with a positive profit is executed (execution decision).
 - The alternative with the higher average profit is preferred (selection decision).
- Sometimes the Return on Investment (RoI) is used as alternative profit comparison method.

Profit Comparison Method – Example

? Which project should be executed?

		Project A	Project B
Capital employed	$i = 9\%$		
	Investment	197,000 €	227,000 €
	Economic lifetime	5 years	5 years
	Liquidation proceeds	12,000 €	17,000 €
Fixed costs p. a.	Maintenance costs	4,000 €	20,000 €
Variable costs p. a.	Material costs	2 € / unit	1.90 € / unit
Sales	Sold units p. a.	9,000 units p. a.	12,000 units p. a.
	Price per unit	10 € / unit	10 € / unit

Profit Comparison Method – Solution

		Project A	Project B
Sales p. a.		9,000 units * 10 € / unit = 90,000 €	12,000 units * 10 € / unit = 120,000 €
	Capital invested	(197,000 € + 12,000 €) / 2 = 104,500 €	(227,000 € + 17,000 €) / 2 = 122,000 €
	Cost of capital	9% * 104,500 € = 9,405	9% * 122,000 € = 10,980 €
	Depreciation cost	(197,000 € - 12,000 €) / 5 = 37,000 €	(227,000 € - 17,000 €) / 5 = 42,000 €
Fixed costs p. a.	Maintenance costs	4,000 €	20,000 €
Variable costs p. a.	Material costs	9,000 units * 2 € / unit = 18,000 €	12,000 units * 1.90 € / unit = 22,800 €
Profit p. a.	Sales - Cost	21,595 €	24,220 €

Profit Comparison Method – Alternative

- The Return on Investment (RoI) is an alternative to decisions based on the average profit p. a.
- The average RoI is the average profit before cost of capital per unit of capital invested:

$$\text{RoI} = \frac{\text{Profit p. a.} + \text{Cost of Capital}}{\text{Capital invested}}$$

- The decision rules for investment projects are:
 - An investment with a RoI above a given hurdle is executed (execution decision).
 - The alternative with the highest RoI is preferred (selection decision).

Profit Comparison Method – Alternative

? Calculate the RoI for projects A and B and interpret the results.

- $\text{RoI(A)} = \frac{21,595 \text{ €} + 9,405 \text{ €}}{104,500 \text{ €}} = 0.2967 = 29.67\%$
- $\text{RoI(B)} = \frac{24,220 \text{ €} + 10,980 \text{ €}}{122,000 \text{ €}} = 0.2885 = 28.85\%$
- In the example project A has a higher RoI, which would lead to the (wrong) decision “Investment A”.
- The RoI is a relative measure that only indicates the profitability of the invested capital, but not the absolute return on the total capital. It can therefore lead to wrong decisions.

Profit Comparison Method – Assessment

- The profit comparison method is only applicable if projects have the same time horizon, as it makes a clear difference whether an average profit is earned only once or more often.
- It does not respect timing differences.
- The preferableness of an investment can depend on the number of produced and sold units (which is easy to manipulate).
- The classification of earnings and costs to a concrete investment project in praxis is anything but easy.
- The simplified calculation of capital invested does (like in cost comparison method) not reflect the real amount of capital employed.
- The Return on Investment does not respect the amount of invested capital.

Payback Period Method – Description

- The payback period method, also known as amortisation method, calculates the time until the capital employed by the initial investment is paid back.
- It assumes that investors prefer fast paybacks of invested capital.
- The decision rules are:
 - An investment with a payback period below an a priori given hurdle is executed (execution decision).
 - The alternative (out of two) with the lower payback period is preferred (selection decision).
- There are two different methods to calculate payback period, the average method and the cumulative method.

Payback Period – Average Method

- The average method calculates the payback period by dividing the initial investment less liquidation proceeds by an average “cash flow p. a.”.
- The average “cash flow p. a.” is calculated by adding the depreciation cost (sometimes also the cost of capital) to the average profit.

$$\text{Payback period} = \frac{\text{Investment} - \text{Liquidation proceeds}}{\text{Average profit p. a.} + \text{Depreciation}}$$

- This correction is needed as depreciation (as well as cost of capital) is not a cash outflow, which means that the actual cash flow is higher than the calculated average profit.

Average Method – Example

? Calculate the payback period for projects A and B.

		Project A	Project B
	Investment	197,000 €	227,000 €
	Liquidation proceeds	12,000 €	17,000 €
	Depreciation cost	37,000 €	42,000 €
Profit p. a.	Sales - Cost	21,595 €	24,220 €

$$\text{PBP (A)} = \frac{197,000 \text{ €} - 12,000 \text{ €}}{21,595 \text{ €} + 37,000 \text{ €}} = 3.16$$

$$\text{PBP (B)} = \frac{227,000 \text{ €} - 17,000 \text{ €}}{24,220 \text{ €} + 42,000 \text{ €}} = 3.17$$

➤ Project A is preferred as it has the lower payback period.

Payback Period – Cumulative Method

- The cumulative adds all cash flows in the period they occur.
- The moment (year) in which the accumulated cash flows reach or exceed the initial investment, is the payback time.

Year	0	1	2	3	4	5
Cash flow	-1,000 €	300 €	300 €	500 €	200 €	400 €
Accumulated cash flows	0	300 €	600 €	1,100 €	1,300 €	1,700 €

➤ The payback period is three (3) years.

Cumulative Method – Example

- An project with an expected lifetime of 10 years has an investment of 200,000 €.
 - In year 1 the cash flow is 50,000 €, it is decreasing every year for 5,000 €.
- ? Calculate the payback period.

Year	0	1	2	3	4	5
Cash flow	-200 €	50 €	45 €	40 €	35 €	30 €
Accumulated cash flows	0	50 €	95 €	135 €	170 €	200 €

✓ The payback period is five (5) years.

Payback Period Method – Assessment

- The average method does not respect timing differences.
- The cumulative method takes different points in time into account, but does not respect that cash flows in future have a different value than cash flows today.
- For execution decisions a time benchmark is set individually by the investor. This is theoretically not acceptable.
- The payback period time is not adequate as it does not take cash flows after the payback period into account. Example:

Year	0	1	2	3
Project A	-1,000 €	1,000 €	100 €	0 €
Project B	-1,000 €	500 €	500 €	1,000 €

Project B???

Dynamic Methods – General Description

- Dynamic methods can be used to calculate the absolute advantage of an investment as well as the relative advantage of (two or more) different investment projects.
- The calculation of the project(s) is based on financial figures (cash flows) rather than on accounting figures (earnings and expenses).
- Dynamic methods do consider the time value of money, i. e. not only the cash flow but also the point in time is respected.
- The crucial criteria for choosing the “right” decision (based on the right capital budgeting method) are the projection of cash flows, the time horizon and not surprisingly the interest rate.

Cash Flow Projection

- There are two ways to calculate future cash flows, the direct and the indirect method.
- The (seldom used) direct method estimates cash inflows and cash outflows directly as they are planned and occur and calculates the net result. As corporate information systems typically do not contain all relevant cash related data, this is extremely difficult.
- The indirect method starts with the net profit and adjusts it for all cash inflows and cash outflows that are not cash effective (e. g. depreciation) and considers additional cash inflows and cash outflows that are not accounted in the income statement (e. g. dividend payments).
- Theoretically both methods must result in the same cash flow.

Cash Flow Projection – Indirect Method

Net Profit

+ Depreciation

+/- Increase/Decrease in provisions

-/+ Changes in working capital

-/+ Changes in non-current assets

+/- Changes in debt

+/- Changes in equity

= Cash Flow

Net Present Value Method

- The Net Present Value method valuates investments based on their future (positive and negative) cash flows.
- If used for execution decisions: Every alternative with a positive NPV is executed, alternatives with negative NPV are not conducted.

$$\text{NPV} = -\text{CF}_0 + \sum_1^n \frac{\text{CF}_t}{(1+i)^t} > 0 \quad \text{or} \quad \sum_1^n \frac{\text{CF}_t}{(1+i)^t} > \text{CF}_0$$

- If used for selection decisions: The alternative (out of two) with the higher NPV is preferred.

Net Present Value Method – Example

- A project with the following cash structure can be acquired, $i = 8\%$:

Year	0	1	2	3	4	5
Cash flow	-260 €	+20 €	+30 €	+50 €	+70 €	+200 €

- ? What is the net present value? Would you invest?
- ? Which debt could be raised in $t=0$? How would a complete finance plan look like?
- ? Which debt could be raised in $t=3$? How would a complete finance plan look like?

Net Present Value Method – Solution

- NPV = $-260 + 20/1.08^1 + 30/1.08^2 + 50/1.08^3 + 70/1.08^4 + 200/1.08^5 = 11,50$

Year	0	1	2	3	4	5
Investment	-260.00	+20.00	+30.00	+50.00	+70.00	+200.00
Loan (+increase/-decrease)	+271.50 → -293.22					
Loan (+increase/-decrease)		+273.22 → -295.08				
Loan (+increase/-decrease)			+265.08 → -286.28			
Loan (+increase/-decrease)				+236.28 → -255.19		
Loan (+increase/-decrease)					+185.19 → 200.00	
Loan (+increase/-decrease)						0
Total	+11.50	0	0	0	0	0

Using the NPV for decisions

- Based on the PV and NPV formulas financial decisions can be made in a comprehensible manner:
 - Projects with an $NPV > 0$ will be executed
 - Projects with an $NPV < 0$ will not be executed.
- This decision rule is applicable for all kind of investments, i. e. real assets and financial securities.
- Nevertheless the understanding of the NPV method in praxis (for non-economists) is partially difficult.
- Therefore other methods can/might/will be applied.
- The NPV rule has also (indirect) implications on financing decisions as it can be used to show that these decisions can be separated from the investment decisions, i. e. done separately.

Internal Rate of Return – Description

- The Internal Rate of Return (IRR) method calculates the relative return that exists implicitly in a project.
- The IRR is the discount rate r where the NPV is equalling 0:

$$\text{NPV} = -\text{CF}_0 + \sum_{t=1}^n \frac{\text{CF}_t}{(1+r)^t} = 0$$

- The decision rules are:
 - If the IRR is above a given hurdle ($r > i$) then a project should be executed.
 - Out of two projects the one with the higher IRR should be chosen (of course only, if the IRR is above a given hurdle).
- In a one-period or two-period-model the IRR can be calculated easily, if $t > 2$ either IT-systems, an approximation or sophisticated mathematical methods must be used.

Internal Rate of Return with t = 1 Period

	0	1
Project A	-157 €	+162 €

$$r = \frac{CF_1}{CF_0} - 1$$

- ? Calculate the IRR.
- ? Decide whether to invest or not with $i = 3\%$
- $157 * (1+r) = 162 \Rightarrow r = 3.18\% \Rightarrow \text{Invest!}$

Internal Rate of Return with t = 2 Periods

	0	1	2
Project A	-240 €	+120 €	+160 €

? Calculate the IRR and decide whether to invest @ $i = 8\%$

$$-240 + 120 / (1+r)^1 + 160 / (1+r)^2 = 0 \quad || \cdot (1+r)^2$$

$$\Rightarrow 240 \cdot (1+r)^2 - 120 \cdot (1+r)^1 - 160 = 0 \quad || :240$$

$$\Rightarrow 1 \cdot (1+r)^2 - 0.5 \cdot (1+r)^1 - 0.67 = 0 \quad || (1+r) = k$$

$$\Rightarrow k^2 - 0.5k - 0.67 = 0$$

$$\text{for } k^2 + a \cdot k + b = 0 \Rightarrow k_{1/2} = -\frac{a}{2} \pm \sqrt{\frac{a^2}{4} - b}$$

$$k_1 = 0.25 + \sqrt{(0.25)^2 + 0.67} \text{ and } k_2 = 0.25 - \sqrt{(0.25)^2 + 0.67}$$

$$k_1 = 1.1039 \Rightarrow r = 10.39\% \text{ and } k_2 = -0.6039 \Rightarrow r = -160.39\%$$



Internal Rate of Return with $t = 2$ Periods

Project	0	1	2
A	-480 €	+50 €	+550 €
B	-500 €	+40 €	+550 €
C	-680 €	+700 €	0 €

? Which of the projects would you chose with $i = 10\%$?

Project	IRR	0	1	2
A	12.38%	-480 €	+50 €	+550 €
B	8.96%	-500 €	+40 €	+550 €
C	2.94%	-680 €	+700 €	0 €

Internal Rate of Return with $t > 2$ Periods

- If $t > 2$ periods, the simplification cannot be used.
- In this case r can only be calculated either with a computer or approximated with a step-by-step method (the so called “regula falsi”):
 1. A linear approximation of r is done by calculating two NPVs with two different interest rates i , one NPV should be positive, the other negative.
 2. In the next step the two NPV- i -combinations are visualised in a diagram and a (linear) NPV (i) -function can be calculated.
 3. Now r can be approximated by setting the NPV (r) = 0.

Internal Rate of Return with $t > 2$ Periods

	0	1	2	3
Project A	-100 €	+70 €	+50 €	+10 €

- $$\text{NPV } (i=15\%) = -100 + 70/1.15^1 + 50/1.15^2 + 10/1.15^3 = 5.25$$
$$\text{NPV } (i=25\%) = -100 + 70/1.25^1 + 50/1.25^2 + 10/1.25^3 = -6.88$$
- With P1 (15%/5.25) and P2 (25%/-6.88) the NPV-interest-function $\text{NPV } (i) = a + b * i$ can be calculated:
$$b = (-6.88 - 5.25) / (25\% - 15\%) = -121.32$$
$$a - 121.32 * 15\% = 5.25 \Rightarrow a = 23.45$$
- $$\text{NPV } (r) = 23.45 - 121.32 * r = 0 \Rightarrow r = 19.33\%$$

Regula falsi – General Formula

$$b = [\text{NPV}(i_2) - \text{NPV}(i_1)] / (i_2 - i_1)$$

$$\text{with } \text{NPV}(i_1) = a + b * i_1 \Rightarrow a = \text{NPV}(i_1) - b * i_1$$

$$\Rightarrow \text{NPV}(i) = \text{NPV}(i_1) - [\text{NPV}(i_2) - \text{NPV}(i_1)] / (i_2 - i_1) * i_1 + [\text{NPV}(i_2) - \text{NPV}(i_1)] / (i_2 - i_1) * i$$

and for $\text{NPV}(r) = 0$:

$$\text{NPV}(r) = 0 = \text{NPV}(i_1) - [\text{NPV}(i_2) - \text{NPV}(i_1)] / (i_2 - i_1) * i_1 + [\text{NPV}(i_2) - \text{NPV}(i_1)] / (i_2 - i_1) * r$$

$$\Rightarrow \mathbf{r} = \mathbf{i_1} - \frac{i_2 - i_1}{\text{NPV}(i_2) - \text{NPV}(i_1)} * \mathbf{NPV(i_1)}$$

Internal Rate of Return with $t > 2$ Periods

	0	1	2	3
Project A	-100 €	+70 €	+50 €	+10 €

- Cross-check with Regula falsi – General formula

$$\text{NPV } (i=15\%) = -100 + 70/1.15^1 + 50/1.15^2 + 10/1.15^3 = 5.25$$

$$\text{NPV } (i=25\%) = -100 + 70/1.25^1 + 50/1.25^2 + 10/1.25^3 = -6.88$$

$$\text{➤ } r = 15\% - [(25\% - 15\%) / (-6.88 - 5.25)] * 5.25 = 19.33\%$$

Internal Rate of Return with $t > 2$ Periods

	0	1	2	3	4	5
Project A	-650 €	+50 €	+150 €	+200 €	+250 €	+300 €
Project B	-300 €	+100 €	+150 €	+130 €		

? Calculate the IRR of both projects based on $i_1 = 10\%$ and $i_2 = 15\%$.

? Which project do you prefer?

	IRR	0	1	2	3	4	5
Project A	11.39%	-650 €	+50 €	+150 €	+200 €	+250 €	+300 €
Project B	12.35%	-300 €	+100 €	+150 €	+130 €		

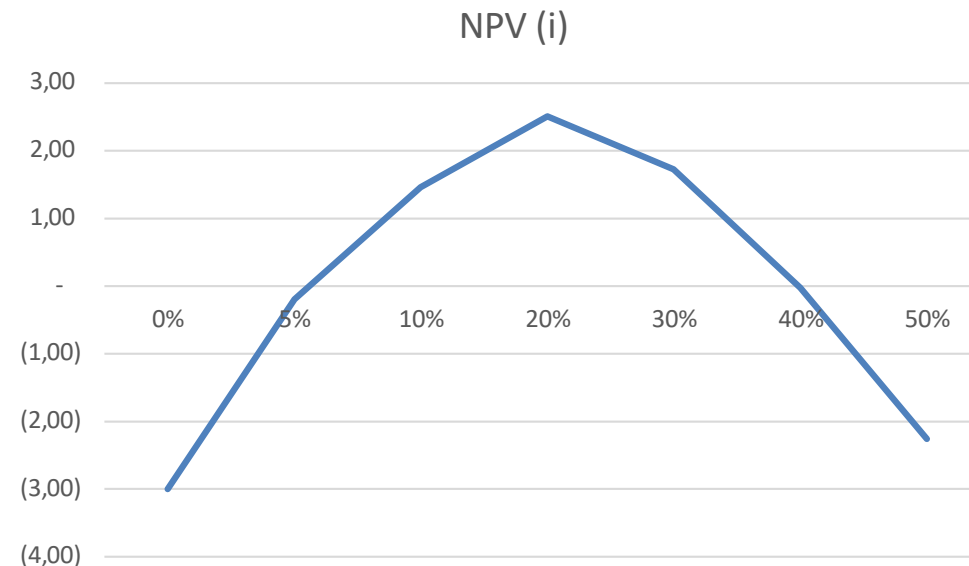
Regula falsi – Assessment

- The Regula falsi assumes a linear correlation between interest and NPV. This is in praxis not (always) valid.
- The closer the two used i for approximation are to r the smaller is the simplification mistake.

Internal Rate of Return – Problems

	0	1	2	3
Project A	-63 €	+80 €	+90 €	-110 €

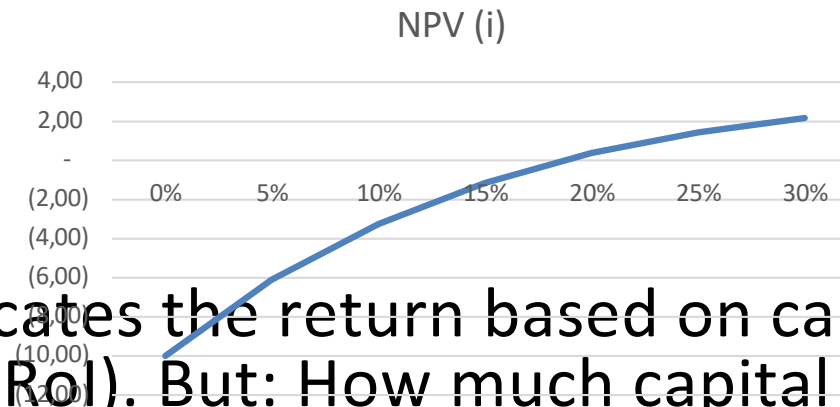
➤ In this case two possible r exist: 5%, 40%



Internal Rate of Return – Problems

	0	1	2	3
Project A	-20 €	+30 €	+ 60 €	-80 €

- In this case the NPV method and the IRR method lead to different results: $i < r \Rightarrow NPV < 0$ and $i > r \Rightarrow NPV > 0$



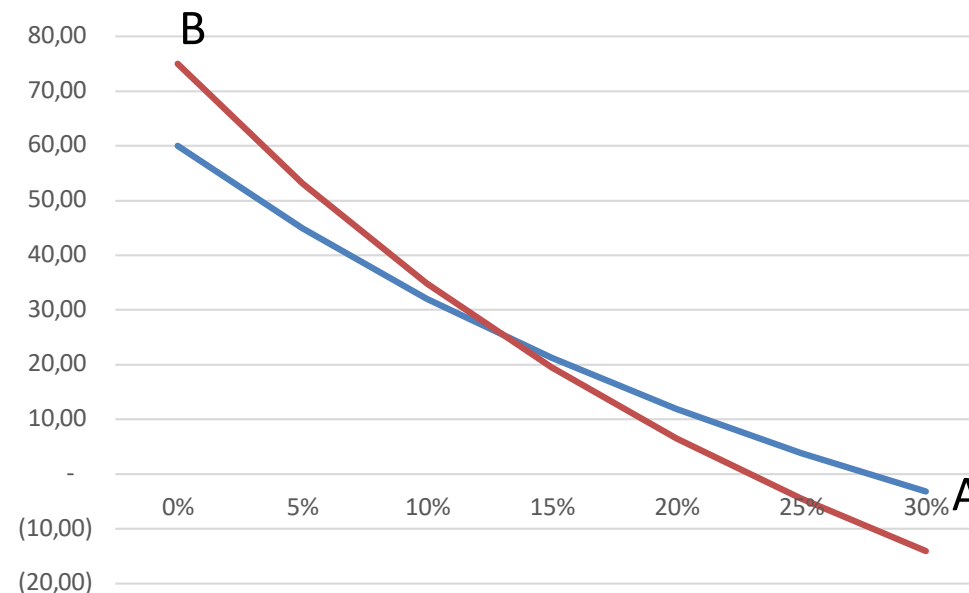
- The IRR indicates the return based on capital invested (as in the static RoI). But: How much capital is invested after year 1?

Internal Rate of Return – Problems

	0	1	2	3
Project A (blue)	-80 €	+20 €	+50 €	+70 €
Project B (red)	-105 €	+20 €	+20 €	+140 €

➤ $i_A (27.05\%) > i_B (22.87\%)$ – Project A is better than project B, but:

➤ But for $0\% < i < 13.02\%$ project B has a higher NPV.



Internal Rate of Return – Assessment

- The IRR is a relative number that indicates the return per unit of capital invested.
- The IRR should only be used with “regular investments” (one cash outflow at the start, then only cash inflows).
- It does not take the total amount of capital invested into account. As this is important for execution decisions, the IRR does not include all relevant information.
- NPV and IRR can lead to different decisions. In this case the NPV is (as always) the right method.
- Nevertheless in praxis the IRR is a frequently used method as it is relatively easy to understand (but not easy to calculate without IT system).

Baldwin Method – General Description

- The Baldwin method calculates the return of an investment (based on cash flows) with focus on two periods t_0 and t_T .
- All cash outflows (= capital invested) are discounted to the PV and added.
- All cash inflows (= capital received) are compounded to the FV and added.
- The return of this “two-period” cash flow is calculated by:

$$CF_0 * (1 + r_B)^T = CF_T \quad \Rightarrow \quad r_B = \sqrt[T]{\frac{CF_T}{CF_0}} - 1$$

- The decision rules are:
 - If the Baldwin interest is above a given hurdle ($r_B > i$) then a project should be executed.
 - Out of two projects the one with the higher Baldwin interest should be chosen (of course only, if the Baldwin interest is above a given hurdle).

Baldwin Method – Example

	0	1	2	3	4
Project A	-80 €	+70 €	-60 €	+40 €	+50 €

- Calculate r_B and decide whether to invest ($i = 7\%$)
- PV (cash outflows) = $-80 - 60/1.07^2 = -132.41$ ($\Rightarrow 132.41$)
- FV (cash inflows) = $70 \cdot 1.07^3 + 40 \cdot 1.07^1 + 50 \cdot 1.07^0 = 178.55$

$$r_B = \sqrt[4]{\frac{178.55}{132.41}} - 1 = 7.76\%$$

$\Rightarrow 7.76\% > 7\% \Rightarrow \text{Invest!}$

Baldwin Method – Examples

	0	1	2	3
Project A	-100 €	+ 60 €	-30 €	+90 €
Project B	-80 €	+50 €	+ 60 €	-19 €

- Calculate r_B and decide whether to invest ($i = 6\%$).
Which project would you prefer?
- Calculate the NPV and decide whether to invest ($i = 6\%$).
Which project would you prefer?

	r_B	NPV
Project A	7.50%	5.47
Project B	7.67%	4.62

Wrong in selection decisions.

Baldwin Method – Assessment

- The Baldwin interest is a kind of “rate of return” and therefore relatively easy to understand.
- Its application is mathematically easy.
- For execution decisions the Baldwin interest leads to the same decision as the NPV method, which is always correct.
- For selection decisions the Baldwin interest might lead to different results than the NPV method, as it is a one period decision rule.

Decisions on the Lifetime of Investments

- The following decisions on the lifetime of investments can be analysed:
 - 1) How long is the ideal useful life of a one-time investment?
 - 2) How long is the ideal useful life of an investment that is replaced once?
 - 3) How long is the ideal useful life of an investment that is replaced infinitely?

Capital Budgeting under Uncertainty

- If future cash flows cannot be predicted with certainty, risk has to be taken into account.
- Possible solutions “to be on the safe side” in capital budgeting methods are
 - Discount on cash inflows
 - Premium on cash outflows
 - Increase interest rate i
- More sophisticated models compute calculations based on risk theory (e. g. CAPM, Bernoulli risk utility function).

Capital Budgeting under Uncertainty

Investment	Year 0	Year 1	Year 2	Year 3
A	- 100	+ 40	+ 40	+ 40
PV (i = 8%)	- 100	+ 37,04	+ 34,29	+ 31,75
PV accumulated	- 100	- 62,96	- 28,67	+ 3,08

Investment	Year 0	Year 1	Year 2	Year 3
PV (i = 9%)	- 100	+ 36,70	+ 33,67	+ 30,89
PV accumulated	- 100	- 63,30	- 29,64	+ 1,25

Investment	Year 0	Year 1	Year 2	Year 3
A	- 100	+ 40	+ 40	+ 35
PV (i = 8%)	- 100	+ 37,04	+ 34,29	+ 27,78
PV accumulated	- 100	- 62,96	- 28,67	- 0,89

Learning Objectives Chapter 3

- ✓ Recapitulate basic formulas for financial tasks
- ✓ Understand the importance of the time value of money
- ✓ Understand purposes, types process and tasks of investments
- ✓ Get to know static methods of capital budgeting,
Apply them to different situations and Assess them
- ✓ Get to know dynamic methods of capital budgeting,
Apply them to different situations and Assess them
- ✓ Calculate lifetime of investments
- ✓ See ways to respect risk in investment decisions

Agenda

1. Fundamentals on the Corporate and on Corporate Finance
2. Fundamentals on Financial Markets
3. Investment Valuation / Capital Budgeting
- 4. Calculation of Capital Needed**
5. Corporate Financing
6. Corporate Analysis
7. International Corporate Finance
8. Derivative Financial Instruments

Learning Objectives Chapter 4

- Understand economic correlations between operative business processes and liquidity needs
- Get to know reasons for capital needed
- Understand parameter for the amount of capital needed
- Learn and Apply how to calculate capital needed in the short-term
- Learn and Apply how to calculate capital needed in the long-term

Content

4. Calculation of Capital Needed

4.1 Fundamentals of the Calculation of Capital Needed

4.2 Calculation of Short-term Capital Needed

4.3 Calculation of Long-term Capital Needed

Reasons for Capital Needed

	Short-term capital needed	Long-term capital needed
Regular reasons	• Payment of supplier invoices • Financing of clients • Pre-financing of growth	• Foundation • Growth • Expansion
Special occasions	• Liquidity shortage	• Corporate transactions / M&A Activities • Restructuring • Re-start after bankruptcy

Determinants of Capital Needed

- Capital needed is influenced by organisational aspects:
 - Strategy
 - Business model
 - Organisation and processes
 - Products, clients, regions
 - ...
- Capital needed is influenced by financial aspects:
 - Existing amount of liquidity (“cash”)
 - Volume, time and probability of cash inflows
 - Volume, time and probability of cash outflows
 - Timing differences between cash inflows and cash outflows

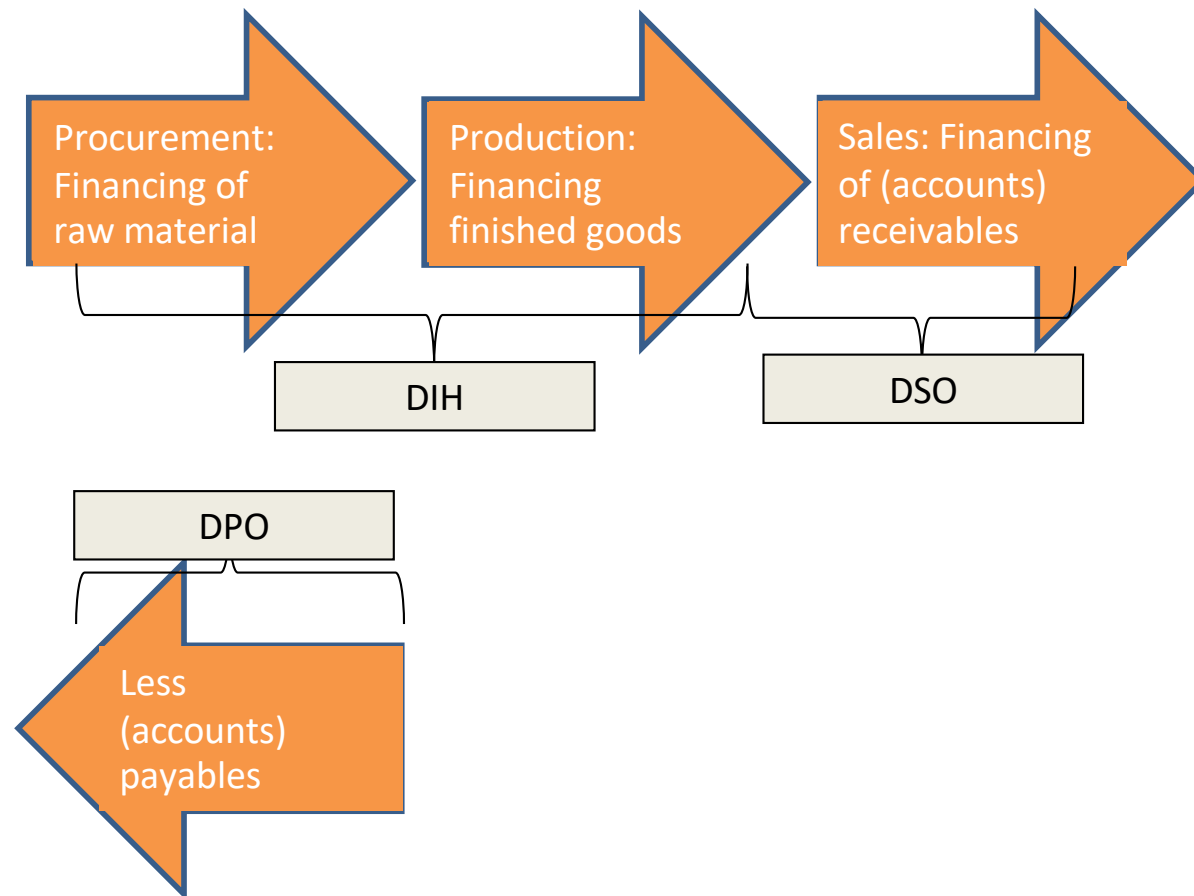
Timing issues liquidity management

- Time horizon: How far do we look into the future?
 - 3 months
 - 6 months
 - 12 months
 - > 12 months
- Planning frequency: How often do we update?
 - daily
 - weekly
 - monthly
 - quarterly

Calculation of Capital Needed

- Non-current assets (“Capital expenditure”, CapEx)
 - Calculation of planned cash outflows and cash inflows
 - based on capital budgeting / investment appraisal
- Current assets (“Working Capital”)
 - Calculation based on the cash conversion cycle (“CCC”)
 - Purchase-to-Pay-Process (“days payables outstanding, DPO”)
 - + Purchase-to-Deliver-Process (“days inventory held, DIH”)
 - + Deliver-to-Cash-Process (“days sales outstanding, DSO”)
 - How long is the capital invested from the cash outflow to suppliers to the cash inflow from clients?

Cash Conversion Cycle

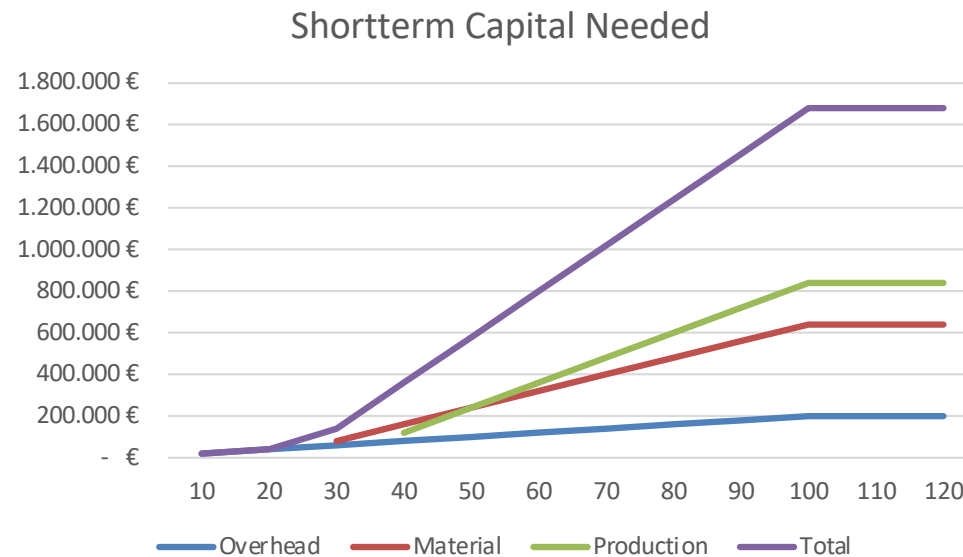


Calculation of Capital Needed – Example

- A company with an existing business has
 - daily material cost of 8,000 €
 - daily manufacturing cost of 12,000 €
 - daily overhead cost of 2,000 €
- Material is stored for 30 days, finished products for 10 days, time needed for manufacturing is 20 days.
- Suppliers offer a 20 day payment period, clients pay 40 days after delivery.
- ? Calculate the total capital needed.

Calculation of Capital Needed – Solution

- Material cost = 8,000 €/day * (30+20+10+40-20) days = 640,000 €
- Manufacturing cost = 12,000 €/day * (20+10+40) days = 840,000 €
- Overhead cost = 2,000 €/day * (30+20+10+40) days = 200,000 €
- Capital needed = 1,680,000 €



Calculation of Capital Needed

- In the long-term, the short-term capital needed must be adjusted for long-term fixation of Bank.
- Additionally finance related cash flows (e. g. dividends, interest payments, repayments) need to be taken into account.
- Calculation based on an integrated corporate planning
 - Income planning (Income statement)
 - Investment planning (Non-current assets)
 - Finance planning (Equity and debt)
 - Cash flow statement

Calculation of Capital Needed – Example

- A company develops a web shop.
 - The expected time to finish the programme is 18 months, the expected costs are 300,000 €, equally distributed.
 - For managing the company and starting the sales, two persons (each @ 30,000 € p. a.) are hired today.
 - General Management and Administration is 20,000 € p. a.
 - The annual marketing expenses starting in year 2 are 30,000 €.
 - Monthly expected turnovers from year 3 onwards are 50,000 €
- ? Calculate the total capital needed and the cash break even.

Calculation of Capital Needed – Example

	Year 1	Year 2	Year 3
Sales	0	0	+ 600,000
R&D cost	- 200,000	- 100,000	0
HR cost	- 60,000	- 60,000	- 60,000
GM&A cost	- 20,000	- 20,000	- 20,000
Marketing cost	0	- 30,000	- 30,000
Liquidity balance (<0 Capital needed >0 capital released)	- 280,000	- 210,000	+ 490,000
➤ Liquidity balance accumulated	- 280,000	- 490,000	+/- 0

➤ The cash break even is after three years.

Learning Objectives Chapter 4

- ✓ Understand economic correlations between operative business processes and liquidity needs
- ✓ Get to know reasons for capital needed
- ✓ Understand parameter for the amount of capital needed
- ✓ Learn and Apply how to calculate capital needed in the short-term
- ✓ Learn and Apply how to calculate capital needed in the long-term

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- 5. Corporate Financing**
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Learning Objectives Chapter 5

- Structure and Explain different forms of financing
- Learn and Apply internal financing with equity and debt
- Learn and Apply external financing with equity
- Learn and Apply external financing with debt
- Understand combinations of financing forms and cost of capital
- See and Apply additional forms of financing

Content

5. Corporate Financing

5.1 Structuration of Financing Forms

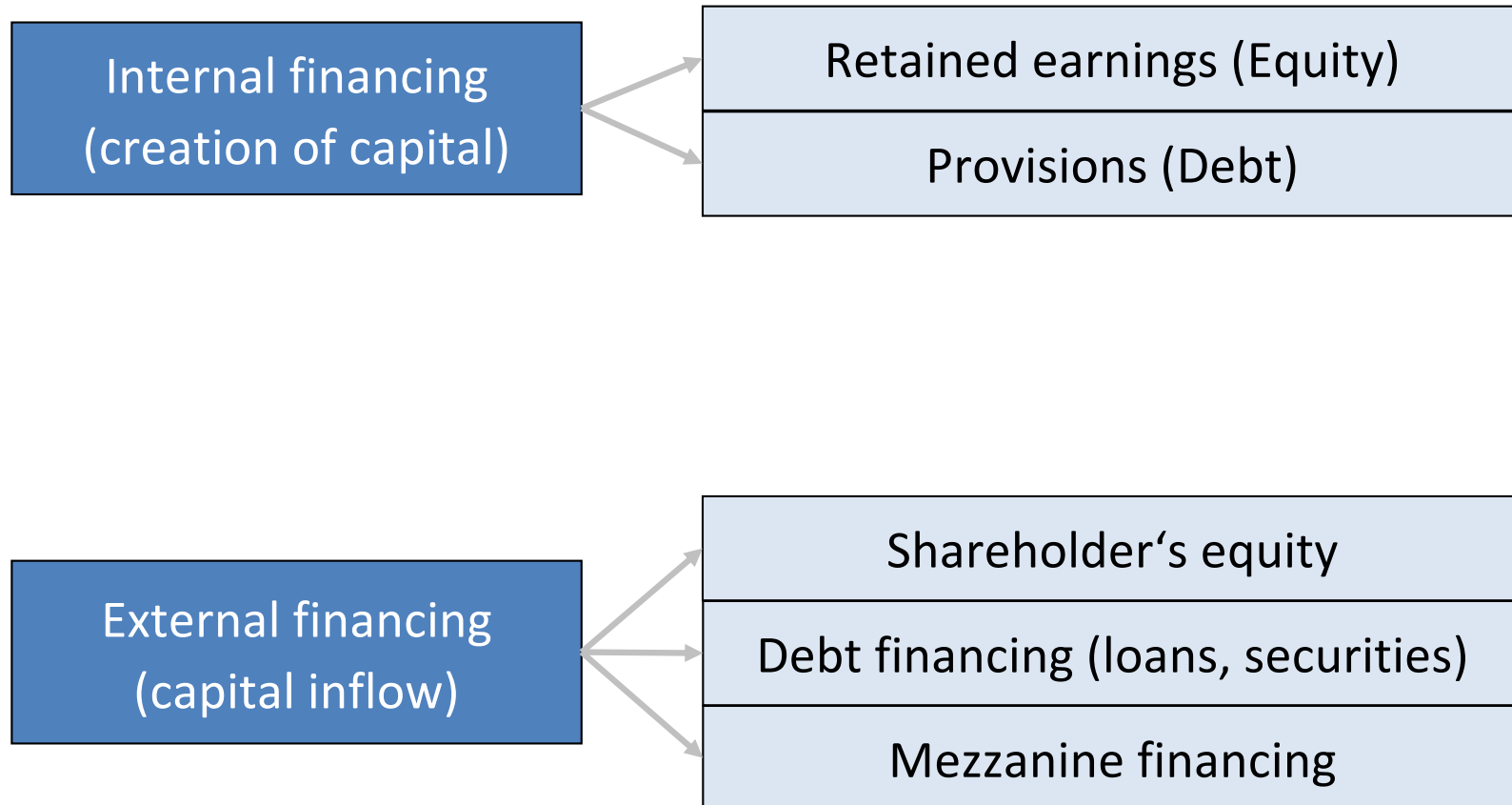
5.2 Internal Financing

5.3 External Financing

5.4 Cost of Capital

5.5 Special Forms of Financing

Forms of Financing – Sources of Funds



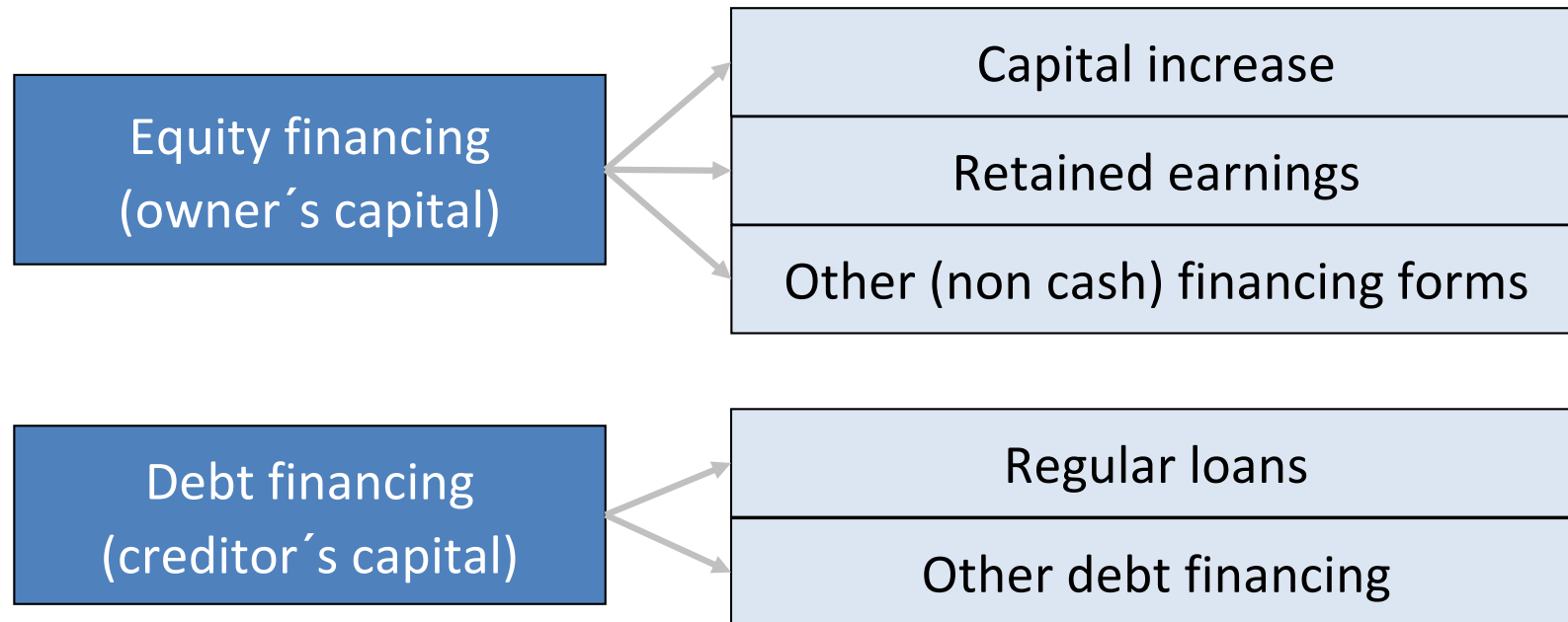
Internal Financing – General Description

- Internal financing (Innenfinanzierung) means the generation or the retaining of liquidity inside the company.
- No liquidity (cash) is inserted into the company from outside the company.
- This financing form can be done either as equity financing (Selbstfinanzierung) or as debt financing (Innenfremdfinanzierung).

External Financing – General Description

- External financing (Außenfinanzierung) means the increase of liquidity from outside the company.
- This financing form can be done either as equity financing (Beteiligungsfinanzierung) or as debt financing (Kreditfinanzierung).
- Mezzanine financing is a mixture between equity and debt financing.

Forms of Financing – Legal Status



Equity and Debt – Main Differences

Criterion	Equity financing	Debt financing
Liability	Fully liable for company's debt Loss carrying stake in company	No liability for company's debt
Profit claims	Claim on net profit	No claim on net profit, but right of fixed payment
Asset claims	Claim on net assets	No claim on assets, but right of preferred repayment
Decision participation	Decision making part of capital	Typically no voting rights
Temporal disposability	Unlimited	Limited
Form of repayment	No fixed/planned repayment	Fixed repayment schedule
Tax treatment	Dividend payments are not tax deductible	Interest payments are tax deductible

Combination of Financing Structures

	Equity financing	Debt financing
Internal financing	Shareholders' capital is built internally. e. g. keeping profits in the company (retained earnings)	Debt capital is built internally. e. g. building provisions for pension plans etc.
External financing	New shareholders' capital is provided to the company. e. g. capital increase	Debt capital is raised externally. e. g. taking a bank loan

Assets	Capital
Non-current assets	Shareholders' capital
	Retained earnings
Current assets	Provisions
	Liabilities

Internal Financing with Equity

- Internal financing with equity means that shareholders' capital is built internally.
- Note that this has only a financing effect if the capital is also liquid, that means not invested in non-liquid assets.
- Internal financing with equity (self financing) can occur in two forms
 - Open self financing (offene Selbstfinanzierung)
 - Silent self financing (stille Selbstfinanzierung)

Open Self Financing

- Open self financing (Offene Selbstfinanzierung) describes the financing of a company via retained earnings.
- This means that profits of the company (which should be liquid) are not distributed to the shareholders (via dividends which reduce liquidity), but kept in the company.
- The financing volume in this case is after tax.

Open Self Financing – Example

Non-current assets	100	Equity	200
Inventories	100		

- Case 1: Inventories are sold for 120 in cash.

Non-current assets	100	Equity	200
Bank	100		
Bank	20	Net profit	20

- Case 2: Inventories are sold for 120 on account.

Non-current assets	100	Equity	200
Receivables	120	Net profit	20

Silent Self Financing

- Silent self financing (Stille Selbstfinanzierung) describes ways to finance a company internally without making the financing process transparent.
- This means that a company generates profits (and as a consequence cash advantages) without accounting these profits in the income statement.
- The generated financing is therefore before tax.
- Examples are
 - Non-recognition or undervaluation of assets (positive stille Reserven)
 - Overvaluation of debt (negative stille Lasten)

Silent Self Financing – Example

Balance sheet according to accounting rules as of 31/12

Non-current assets	100	Equity	100
Bank	100	Debt	100

- The market value of the non-current assets is 150.

“Balance sheet” according to fair market values as of 31/12

Non-current assets	150	Equity	100
Bank	100	Equity („Hidden reserves“)	50
		Debt	100

- It is clear, that silent self financing -if at all- only has indirect financing effects.

Internal Financing via Depreciation

- Financing via depreciation is a typical way to finance a company.
- Main prerequisite is that the accounted (or calculated) depreciation expenses (or cost) can be earned via liquid cash inflows.
- Depreciation will lead to a release of capital and typically also to an increase of capacity.

Depreciation – Release of Capital

- Release of Capital assumes that the value of depreciation is earned in liquidity (via the calculation of depreciation in the sales price) and therefore is available already before a reinvestment needs to be done.
- In that manner non-current assets can be replaced without the need for new capital.
- Example: Time horizon 4 years, each year acquisition of one machine for 8,000 €, expected lifetime = 4 years, straight-line depreciation

Release of Capital – Example

	Increase of capacity				Reinvestment phase			
Year	1	2	3	4	5	6	7	8ff.
M1 (@ 8,000)	2,000	2,000	2,000	2,000	2,000	2,000	2,000	etc.
M2 (@ 8,000)		2,000	2,000	2,000	2,000	2,000	2,000	etc.
M3 (@ 8,000)			2,000	2,000	2,000	2,000	2,000	etc.
M4 (@ 8,000)				2,000	2,000	2,000	2,000	etc.
Depreciation	2,000	4,000	6,000	8,000	8,000	8,000	8,000	etc.
Cash p. a.	2,000	4,000	6,000	8,000	8,000	8,000	8,000	etc.
Σ Cash	2,000	6,000	12,000	20,000	20,000	20,000	20,000	etc.
Reinvestment	0	0	0	8,000	8,000	8,000	8,000	etc.
Capital released	2,000	6,000	12,000	12,000	12,000	12,000	12,000	etc.

Depreciation – Increase of Capacity

- The released liquidity generally is larger than the financing need for replacement investments.
- The released liquidity can therefore be used to invest into additional assets which leads to an increase of capacity (also known as Lohmann-Ruchti-Effect).
- The increase of capacity can be calculated with

$$\text{Increase of capacity} = \frac{2}{1 + \frac{1}{n}}$$

with n = expected lifetime for straight-line depreciation

- Example: A company has 5 machines, each for 4,000 €, Expected lifetime = 4 years, straight-line depreciation

Increase of Capacity – Example

Year	Machines			Cost/Value	Dep.	Σ Cash	Reinvestment and Increase	Capital release
	↑	↓	=					
1	+5	0	= 5	20 T€	5 T€	5 T€	4 T€	1 T€
2	+1	0	= 6	24 T€	6 T€	7 T€	4 T€	3 T€
3	+1	0	= 7	28 T€	7 T€	10 T€	8 T€	2 T€
4	+2	0	= 9	36 T€	9 T€	11 T€	8 T€	3 T€
5	+2	-5	= 6	24 T€	6 T€	9 T€	8 T€	1 T€
6	+2	-1	= 7	28 T€	7 T€	8 T€	8 T€	0 T€
7	+2	-1	= 8	32 T€	8 T€	8 T€	8 T€	0 T€
8	+2	-2	= 8	32 T€	8 T€	8 T€	8 T€	0 T€
9ff.	etc.	etc.	etc.	etc.	etc.	etc.	etc.	etc.

Internal Financing with Debt

- Internal financing with debt means that debt capital is built internally.
- Note that this has only a financing effect if the capital is also liquid, i. e. cash available.

Internal Financing with Provisions

- Financing via provisions is a classical way to finance a company without external funding.
- Typically only long-term provisions with manageability by the company are chosen for financing aspects.
- An -especially in former times- often used form are pension provisions, as they are long-term debt and the structure of the provision can be built according to companies needs.
- The effects of building provisions on financing capacities is based on a timing difference between building the provision and using the provision. Additionally a tax reduction effect occurs.
- The financing effects are nevertheless only temporary as the obligations need to be fulfilled.

Financing with Provisions – Example

Non-current assets	100	Equity	200
Inventories	100		

1. Inventories are sold for 120 in cash:

Non-current assets	100	Equity	200
Bank	120	Net profit	20

2. The company grants a pension benefit of 5:

Non-current assets	100	Equity	200
Bank	120	Net profit	15
		Provisions	5

Financing with Provisions – Example

		Without provision	With provision
	Earnings before tax	500,000 €	500,000 €
-	<u>Increase of provisions</u>	<u>0</u>	<u>100,000 €</u>
=	Taxable income	500,000 €	400,000 €
-	<u>Income tax (30%)</u>	<u>150,000 €</u>	<u>120,000 €</u>
=	Net profit	350,000 €	280,000 €
The financing effects out of building the provision are			
=>	Maximum dividend	350,000 €	280,000 €
=>	Financing potential	350,000 €	280,000 €
		a reduced potential for dividends (as net profit is lower) a reduced tax payment (temporary effect)	
		$280,000 \text{ €} + 100,000 \text{ €} = 380,000 \text{ €}$	

External Financing with Equity

- External financing with equity means that new shareholders' capital is provided to the company from outside.
- Privately owned companies, partnerships and limited liability companies cannot issue equity titles on regulated markets.
- Therefore the possibilities to access (external) financing are limited.
- Corporations do have the access to financial (in this context: capital) markets. (Note that this is also valid for debt capital markets)

Shares/Stock – General Description

- A share (or also a stock) is one unit of ownership interest in a corporation or a financial asset.
- While “shares” are often used to refer to the stock of a corporation, the term shares can also represent ownership of other classes of financial assets, such as mutual funds.
- Owning shares in a business does not mean that the shareholder has direct control over the business's day-to-day operations.
- Being a shareholder does entitle the owner to an equal distribution in any profits and/or dividends as well as liquidation proceeds.

Non Tradable Equity Titles

- Non tradable shares or equity titles (i. e. those that may not be traded in an organised market) are no securities.
- Consequences
 - It is difficult for the company (and existing shareholders) to assimilate new shareholders and therefore difficult to raise new equity.
 - It is difficult for shareholders to sell the shares, so only a limited fungibility of the title is given.
 - It is not possible to objectively determine a fair price for the share.
- Nevertheless there is a market for these kind of participations: www.eigenkapitalforum.com

Tradable Equity Titles

- Tradable equity titles may be listed in an organised market, i. e. at a stock exchange
 - e. g. Deutsche Börse in Frankfurt or NYSE in New York
 - There are also markets
 - for agricultural products such as CME in Chicago
 - for derivative instruments such as EUREX in Frankfurt
 - for electrical energy and emission certificates such as EEX in Leipzig
- Brokers act as market maker, so there is always a market price
 - Stock can be bought or sold at (almost) any time.
 - Nevertheless the market price must not be objective.

Types of Shares in Corporations

- Bearer share (Inhaberaktie) vs. Registered share (Namensaktie)
- Par value share (Nennwertaktie) vs. No-par-value share (nennwertlose Aktie)
- Ordinary share (Stammaktie) vs. Preferred share (Vorzugsaktie)
 - Voting rights => main decision making part of equity
 - Ordinary right for dividends, but obligation to step behind preferred shareholders in case of losses
- Preferred share (Vorzugsaktie)
 - No voting rights => no influence on corporate decisions
 - Preferred rights, typically higher dividend claim

Types of Capital Increase

- Regular capital increase (182-191 AktG)
 - The share capital can be increased via payment in cash.
 - An alternative is the insertion of assets.
- Conditional capital increase (192-201 AktG)
 - The capital will be increased if an a priori defined condition is fulfilled.
 - The maximum amount is 50% of the share capital.
- Authorised capital (202-206 AktG)
 - The management board has the right to issue new shares.
 - Up to 10% can be issued without purchase rights for existing shareholders.
- Capital increase from own resources (207-220 AktG)
 - New share are issued by transforming existing reserves (capital surplus / retained earnings) into share capital.
 - Note that this does not lead to a direct increase in financing potential.

Regular Capital Increase – Description

- Within a regular capital increase the company creates new shares that are issued (i. e. sold) to existing or new shareholders.
- The company has an increase in cash (exception: contribution of assets) and in equity.
- The number of shares issued is depending on the price per share (which depends on the value of the company) and the voting rights being attributed to the new shares.
- The combination of volume of the capital increase and voting rights distribution is an important factor in assessing the effects of a capital increase from company's perspective as well as from shareholders' perspective.
- The loss of influence and profits for the existing shareholders is called dilution.

Regular Capital Increase – Example 1

Non-current assets	100	Shareholding capital	100

- The shareholding capital is split in 100 shares.
- The company issues 100 new shares with a nominal value of 1 € for an issuing price of 1 €.

The new shares are subscribed by a new shareholder:

Non-current assets	100	Shareholding capital „old“	100
Bank	100	Shareholding capital „new“	100

- The existing shareholders' quote is decreasing from (100 shares / 100 shares =) 100% to (100 shares / 200 shares =) 50%, the new shareholders will have (100 shares / 200 shares =) 50% of the voting rights.
- Company value after capital increase is 200 shares * 1€/share = 200 €. Before the capital increase value was 200 € - 100 € = 100 €.

Regular Capital Increase – Example 2

Non-current assets	100	Shareholding capital	100

- The shareholding capital is split in 100 shares.
- The company issues 50 new shares with a nominal value of 1 € for an issuing price of 2 € (i. e. with a premium, called “agio” of 1 €).

The new shares are subscribed by a new shareholder:

Non-current assets	100	Shareholding capital	$(100 + 50 * 1 =) 150$
Bank	100	Paid-in capital	$(50 * 1 =) 50$

- The existing shareholders' quote is decreasing from (100 shares / 100 shares =) 100% to (100 shares / 150 shares =) 66,67%, the new shareholders will have (50 shares / 150 shares =) 33,33% of the voting rights.
- Company value after capital increase is 150 shares * 2€/share = 300 €. Before the capital increase value was 300 € - 100 € = 200 €.

Regular Capital Increase – Example 3

Non-current assets	100	Shareholding capital	100

- The shareholding capital is split in 100 shares.
- The company issues 25 new shares with a nominal value of 1 € for an issuing price of 4 € (i. e. with a premium, called “agio” of 3 €).

The new shares are subscribed by a new shareholder:

Non-current assets	100	Shareholding capital	$(100 + 25 * 1 =) 125$
Bank	100	Paid-in capital	$(25 * 3 =) 75$

- The existing shareholders' quote is decreasing from (100 shares / 100 shares =) 100% to (100 shares / 125 shares =) 80%, the new shareholders will have (25 shares / 125 shares =) 20% of the voting rights.
- Company value after capital increase is 125 shares * 4€/share = 500 €. Before the capital increase value was 500 € - 100 € = 400 €.

Regular Capital Increase – Exercise

- Company C has 40 shares, each with a face value of 1 and retained earnings of 60. It owns assets with a book value of 100.
- It plans to invest 100. The board has decided to finance the investment solely via a capital increase.
- ? Please prepare the balance sheet C before the capital increase.
- ? What is the volume needed for the capital increase?
- ? How many shares would you raise to meet the finance obligation? What will be the price for one share?
- ? What are the effects on existing shareholders, if they participate / do not participate in the capital increase?

Regular Capital Increase – Solution

Before Capital Increase			
Assets	100	Share capital	40
		Ret. earnings	60

← 100%



~~e. g. issue of 40 shares with a face value of 1 for a price of 2.5 each~~

After Capital Increase			
Assets	100	Share capital	$40 + 40 \cdot 1 = 80$
		Ret. earnings	60
Bank	100	Paid-in capital	$40 \cdot (2.5 - 1) = 60$

← 50%

← 50%



of 10 each

After Capital Increase			
Assets	100	Share capital	$40 + 10 \cdot 1 = 50$
		Ret. earnings	60
Bank	100	Paid-in capital	$10 \cdot (10 - 1) = 90$

← 80%

← 20%



Dilution and Purchase Rights

- The increase of capital leads to a loss for the existing shareholders:
 - Loss in voting rights
 - Loss in profit/dividend/wealth
- Existing shareholders therefore are legally granted the right to buy the shares first. This is called purchase right.
- Exception: In cases of conditional capital increase and authorised capital the board can issue new shares without purchase rights for existing shareholders.
- The purchase right shall enable the shareholder to keep his financial and legal position unchanged.
- The purchase right is an independent financial security, so it can be traded individually.

Capital Increase – Formulas

- Subscription ratio (Bezugsverhältnis):

$$\text{Subscription ratio} = \frac{\text{Number of old shares}}{\text{Number of new shares}}$$

- Value of the purchase right (Bezugsrecht):

$$\text{Purchase right} = \frac{\text{Price old share} - \text{Price new share}}{\text{Subscription ratio} + 1}$$

- Share price after capital increase (rechnerischer neuer Aktienkurs):

$$\text{Share price} = \frac{\text{Existing equity} + \text{New equity}}{\text{Number old shares} + \text{Number new shares}}$$

OR

$$\text{Share price} = \text{Price old share} - \text{purchase right}$$

Purchase Rights – Example

- A Company has 60 Mio. shares. The share price is 10 €/share.
- It plans to issue 20 Mio. new shares, the planned issuing price is 8 €/share.
- ? Calculate the subscription ratio, the value of one purchase right and the new calculated share price.
- ? Shareholder S owns 30 shares and 1,000 € in cash. What are his possibilities when he wants to or does not want to participate in the capital increase?
- ? What financial consequences result in his decisions?

Purchase Rights – Solution

- Subscription ratio = 60 Mio. / 20 Mio. = 3
- Purchase right = $(10 \text{ €} - 8 \text{ €}) / (3+1) = 0.50 \text{ €}$
- New (calculated) share price = $(60 \text{ Mio.} * 10 \text{ €/share} + 20 \text{ Mio.} * 8 \text{ €/share}) / (60 \text{ Mio.} + 20 \text{ Mio.}) = 9.50 \text{ €}$

OR

- New (calculated) share price = $10 \text{ €} - 0.50 \text{ €} = 9.50 \text{ €}$

Purchase Rights – Solution

Wealth (start): $30 \text{ shares} * 10 \text{ €/share} + 1,000 \text{ €} = 1,300 \text{ €}$

1. Participating in capital increase: Acquisition of 1 new share per 3 old shares => 10 new shares for total 80 €.

➤ Wealth: $40 \text{ shares} * 9.50 \text{ €/share} + 920 \text{ €} = 1,300 \text{ €}$

2. Not participating in capital increase: Sale of all 30 purchase rights for a total of 15 €.

➤ Wealth: $30 \text{ shares} * 9.50 \text{ €/share} + 1,015 \text{ €} = 1,300 \text{ €}$

3. Partly participating in capital increase: Sale of 15 purchase rights for a total of 7,50 € and acquisition of 5 new shares for total 40 €.

➤ Wealth: $35 \text{ shares} * 9.50 \text{ €/share} + 967.50 \text{ €} = 1,300 \text{ €}$

Reasons for Going Public

- Company related reasons
 - Create sufficient equity basis
 - Increase liquidity
 - Independence from owner's financial strength
 - Reduce capital cost
 - Positive image effect
 - „Bandwagon effect“: Competitors go public, so we also have to
- Shareholder related reasons
 - Going concern and continuity of life's work
 - Control of management
 - Risk diversification through numerous investors

Maturity to Going Public

- The maturity of a corporation to go public depends (beneath fulfilment of objective legal requirements) on many (partly subjective) criteria.
- Some aspects are
 - Sales (total) and sales growth rate
 - Present and future profitability
 - Cashflow and future dividends
 - Strategic/competitive situation
 - Quality of top level management
 - Organisational structure and processes
 - Business model
 - Internal and external accounting

Disadvantages of Going Public

- Cost directly related to the IPO process
- Accounting and publicity
 - Report for admission to stock exchange
 - Sales prospectus
 - General accounting requirements
 - Transparency
 - Financial statements and intermediate reporting
 - Ad-hoc obligations
 - Regulations on insider trading
- Loss of autonomy/self control
- Tax laws

Other Forms of Equity

- Private equity financing
- Venture capital financing
 - Seed financing
 - Start-Up financing
 - First stage financing
 - Bridge financing
- Business Angel financing
- Buy-Out Financing
 - Management Buy-out (MBO)
 - Management Buy-in (MBI)
 - Leveraged Buy-out (LBO)

External Financing with Debt

- External financing with debt means that new debt capital is raised from outside the company.
- Debt is a financing form that has to be paid back in a given timeframe.
- Debt providers follow different purposes based on the type of debt that they grant.
- Debt capital can be structured based on
 - Debt provider (financial institution, supplier, private person)
 - Time horizon of the debt
 - Type of security offered by the borrower

Debt Financing – Purposes

- Quantitative reasons
 - Increase return on equity: If the return on total capital is higher than the cost for debt, additional debt will increase the (relative) return on equity (“leverage effect”).
 - Increase investment volume: Due to potential limitations in equity additional projects can only be handled with additional (debt) finance.
 - Tax optimisation: As interest payments are tax deductible, debt finance can help to optimise the taxable income.
- Qualitative reasons
 - With debt financing the shareholder structure is not affected (negatively), i. e. decision rights remain unchanged.

The Leverage Effect – Formula

- If $\text{RoCE} > i$, additional Debt increases the RoE:

$$\text{RoE} = \text{RoCE} + (\text{RoCE} - i) * \frac{\text{Debt}}{\text{Equity}}$$

with

RoE = Return on equity

RoCE = Return on capital employed

i = Interest rate for debt

Debt / Equity = Gearing / Leverage

- Note that this is only valid on absolute numbers.

The Leverage Effect – Example

i=10%	RoCE = 12%	RoCE = 6%
Debt / Equity = 0 / 1,000 = 0	RoE = 12%	RoE = 6%
Debt / Equity = 500 / 500 = 1	RoE = 14%	RoE = 2%
Debt / Equity = 900 / 100 = 9	RoE = 30%	RoE = -30%

- It is obvious that the leverage effect can turn to a leverage risk!

Increase of investment volume

Investment without debt		Investment with debt	
Assets 100	Equity 100	Assets 500	Equity 100
			Debt 400

- Increasing the investment volume leads to an increase in the RoE:

Income Statement		Income Statement	
RoCE (assumption 5%)	5	RoCE (assumption 5%)	25
<u>- Interest</u>	<u>0</u>	<u>- Interest (assumption 4% on debt)</u>	<u>- 16</u>
= Earnings before taxes	5	= Earnings before taxes	9
Return on equity	5%	Return on equity	9%

Tax advantage of debt capital

Financing with equity		Financing with debt	
Assets 250	Equity 250	Assets 250	Debt 250

- Different forms of finance lead to different results:

Income Statement		Income statement	
EBIT (Earnings before interest + taxes)	20	EBIT (Earnings before interest + taxes)	20
- Interest	0	- Interest (assumption 4% on debt)	- 10
= Earnings before taxes	20	= Earnings before taxes	10
- Taxes (assumption 30%)	- 6	- Taxes (assumption 30%)	- 3
= Net profit	14	= Net profit	7
Financing with equity leads to higher profits.			
Net profit + Interest	14	Net profit + Interest	17

But financing with debt leads to higher total results (profit + interest) as interest payments lead to a reduction in tax ("tax shield").

Credit Decision of the Creditor (Bank)

- Quantitative credit scoring („hard“ facts)
- Qualitative credit scoring („soft“ facts)
- Availability (and result) of an external rating
- Profit of the bank from granting the loan
- Collateral / Securitisation
- Covenants

Quantitative Factors of Credit Worthiness

- Financial statement analysis and analysis of financial plans
 - Information on assets and liabilities, profit/loss, cash flow
 - Time series analysis (3 to 5 years past and future)
 - Industry analysis (benchmarks)
- Important ratios
 - Debt/equity ratio
 - Equity-to-fixed-assets ratio
 - Liquidity ratios
 - Return on Assets (ROA), Return on Equity (ROE)
 - Working capital ratios
 - Interest Cover Rate (ICR), Debt Service Cover Rate (DSCR)
- Depending on legal form: private wealth of the debtor

Qualitative Factors of Credit Worthiness

- Information on previous payment behaviour
 - Account analysis (for known customers)
 - Third party information (bank inquiry, agency reports – e. g. Schufa, large-scale credit information from central banks)
- Information on operating business of the company
 - Annotations to financial statements (e. g. on valuation methods)
 - Assessment of the specific project to be financed
- Information on market environment
- Information on organisation and processes
 - Accounting and controlling
 - Logistics, production, technology
 - Owner and management
 - Staff and organisation

Rating Classes S&P, Moody's, Fitch

S&P	Moody's	Fitch	Denotation	Description
AAA	Aaa	AAA	Prime (Triple A)	Highest rating, almost no default risk
AA+	Aa1	AA+	High grade	High probability of payments, low default risk
AA	Aa2	AA		
AA-	Aa3	AA-		
A+	A1	A+	Upper Medium grade	Adequate payment of interest and repayments; existence of risk factors that might have negative impacts if corporate environment is changed
A	A2	A		
A-	A3	A-		
BBB+	Baa1	BBB+	Lower Medium grade	Adequate payment of interest and repayments; speculative elements or missing securitisation against changes in corporate environment
BBB	Baa2	BBB		
BBB-	Baa3	BBB-		

Rating Classes S&P, Moody's, Fitch

S&P	Moody's	Fitch	Denotation	Description
BB+	Ba1	BB+	Non Investment grade speculative	Poor payment of interest and repayments, even in good economic conditions
BB	Ba2	BB		
BB-	Ba3	BB-		
B+	B1	B+	Highly Speculative	Low payment of interest and repayments
B	B2	B		
B-	B3	B-		
CCC+	Caa1	CCC+	Substantial risks / Extremely speculative	Lowest quality of investments, lowest securities for investors, high danger of default
CCC	Caa2	CCC		
CCC-	Caa3	CCC-		
CC	Ca	CC		
C	C	C	In default	In default
D		D		

Securities – Definition and Functions

- Securities are guarantees in loan agreements that shall secure the loan and ensure the payment of interest and the complete repayment of the loan.
- Securities are linked to the loan / to an asset / to the debtor.
- Loan/Asset related securities are
 - Reservation of proprietary rights (Eigentumsvorbehalt)
 - Mortgages (Hypotheken)
 - Land charges (Grundschulden)
 - Pledges (Pfandrecht)
- Debtor related securities are
 - Guaranties (Bürgschaft, Garantieerklärung)
 - Comfort letter (Patronatserklärung)

Covenants – Definition and Functions

- Covenants are clauses in loan agreements that shall prevent the loan from default by putting restrictions on the borrower.
- These clauses are to be fulfilled by the borrower, otherwise the lender has the right for an extraordinary termination of the loan contract.
- In this case the borrower has to insert additional equity into (i. e. reduce the loan amount) or even repay the outstanding balance of the loan.
- Covenants can be financial as well as non-financial.

Short-term Financing with Debt

- Accounts payables: If suppliers receive their payments with a time lag, the company uses external debt financing (typically at low cost). If however the supplier offers a discount for early payment, this financing form can become expensive.
- Another possibility (that is not at low cost either) is short-term bank financing with a current account (Kontokorrentkonto).
- Both forms are frequently seen, but should only be used for short-term cash needs.

Short-term Financing with Debt

- Calculation interest cost for a trade discount:

$$i = \frac{\text{discount rate} * 360}{\text{due date for payment} - \text{discount period}}$$

- Example: A supplier offers a discount of 2% for a payment with 10 days. The regular payment time is 30 days.

$$i = \frac{2\% * 360}{30 - 10} = 36\% \text{ (p. a.)}$$

Short-term Financing with Debt

- Calculation of interest cost for a current account with interest settlements during the year:

$$i_{\text{eff}} = \left(1 + \frac{i_{\text{nom}}}{n}\right)^n - 1$$

- Example: A bank settles the outstanding balance of a cash account every quarter. The nominal interest rate is 10% p. a.

$$i_{\text{eff}} = \left(1 + \frac{0,10}{4}\right)^4 - 1 = 10.38\%$$

Financing with Loans – Characteristics

- A loan is an individual contract between the provider of capital/investor (creditor, often a bank) and capital seeker (debtor, natural persons or companies).
- Loan financing is a form of external debt financing.
- Debt financing is characterised by
 - There is no (direct) liability for other corporate debt.
 - The return (interest) is independent from corporate success.
 - The time frame for granted capital is limited (maturity of the loan, repayment of the outstanding principal).
 - Creditors do only have information rights or control rights, but no decision rights.

Financing with Loans – Characteristics

- Interest rate
 - fixed
 - flexible
 - individual agreement (e. g. profit sharing)
- Repayment (Amortisation)
 - fixed
 - flexible
 - only at termination (e. g. “bullet payment”)
 - extraordinary repayments
- Duration
 - fixed termination
 - flexible duration

Financing with Loans – Characteristics

- Additional loan fees and borrowing cost
 - Securitisation
 - Land charge
 - Mortgage
 - Syndication
- Covenants
 - Financial covenants
 - Non-financial covenants
 - Information covenants
- Liability of borrower
 - Full-recourse
 - Non-recourse

Loans – The Lender's view

- The lender needs to refinance his loan investment. The cost for refinancing are typically below the earnings on the loan.
- A main task for the lender is to coordinate the timing of the loan and his refinancing agreement(s).
- The margin for the lender is frequently measured in basis points (“bps”), with 100 basis points = 1%.
- The margin leads to profit for the lender, but must also cover administration cost and losses (defaults) on loans.
- Additional cost can be invoiced, such as appraisal fees, underwriting fees, forward rate agreement fees, etc.

Loans – The Lender's view

Lender/Bank (before loan and refinancing)				
Assets	50		Equity	10
			Debt	40

Lender/Bank (after loan and refinancing)				
Assets	50		Equity	10
			Debt	40
Loan (@6%)	100		Refinancing (@4%)	100

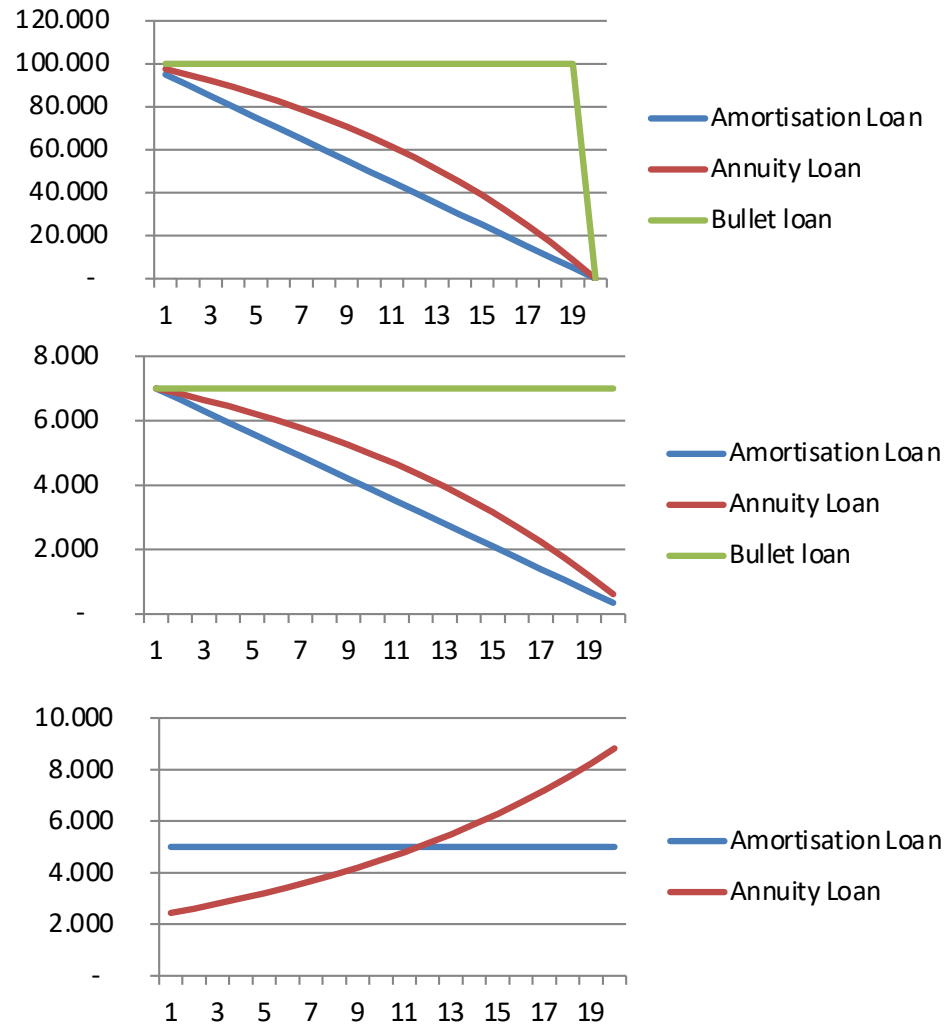
Lender/Bank (annual income statement)				
<u>Cost for refinancing</u>	<u>4</u>		Interest income	6
Profit	2			

Amortisation Schedules for Loans

- Amortisation free loan (Bullet loan)
 - Until maturity only interest payments
 - Full repayment at maturity
- Constant amortisation loan (CAL) / Constant repayment loan
 - Linear amortisation of loan amount
 - Decreasing total payments due to decreasing interest payments
- Constant payment loan (CPL) / Annuity loan
 - Regular payment structure with fixed amount
 - Interest payments decrease whereas amortisation increases
 - In praxis (e. g. mortgage loans) the initial amortisation frequently is given, e. g. „initial amortisation 1% p.a.“
- Balloon repayment
 - Until maturity no payments at all
 - Full repayment (including accumulated interest) at maturity

Long-term Loans – Differentiation

- Outstanding loan balance
- Interest payments
- Repayments



Constant Amortisation Loan

- Within a constant amortisation (CAL) a regular debt repayment is calculated and kept constant during duration of the loan agreement.
- As interest is calculated on outstanding loan balance, the total payment is decreasing.
- Example: A loan of 100,000 € @ 5% interest is paid back in 5 years. Calculate the amortisation for one month and the interest in month 1, month 13 and month 49.
 - Amortisation: $100,000 \text{ €} / (5 * 12) = 1,666.67 \text{ €}$ per month
 - Interest for month 1: $5\% * 100,000 \text{ €} / 12 = 416.67 \text{ €}$
 - Interest for month 13: $5\% * (100,000 \text{ €} - 12 * 1,666.67 \text{ €}) / 12 = 333.33 \text{ €}$
 - Interest for month 49: $5\% * (100,000 \text{ €} - 48 * 1,666.67 \text{ €}) / 12 = 83.33 \text{ €}$

Constant Payment Loan / Annuity Loan

- Within a constant payment loan (CPL) a constant payment loan has a fixed payment that is kept constant during duration of the loan agreement.
- This is also called annuity loan (Annuitätendarlehen).
- As interest is calculated on the outstanding loan balance and as the total payment remains constant, the interest payments are decreasing and the debt repayments are increasing.
- The lower the interest rate, the lower is the replacing repayment amount.
- CPL can be structured as a full repayment loan or a part repayment loan. In the second case at the end of the duration an outstanding loan balance needs to be settled or prolonged.

Constant Payment Loan – Formulas

- Calculation of the annual payment a :

$$a = d * \frac{i * (1 + i)^n}{(1 + i)^n - 1}$$

with d = debt, n = years, i = interest rate

- Calculation of the total interest cost:

$$\text{total interest paid} = n * a - d$$

- Calculation of the duration of a full repayment loan:

$$n = \frac{\ln(i + r) - \ln(r)}{\ln(1 + i)}$$

- Calculation of the outstanding loan balance after m years:

$$d * \frac{(1 + i)^n - (1 + i)^m}{(1 + i)^n - 1}$$

Amortisation Schedules – Examples

- Develop the payment and amortisation schedule for a loan with loan principal 100,000 EUR, interest 5%, time to maturity 5 years as a
 - bullet loan
 - constant repayment loan
 - annuity loan
- Develop the payment and amortisation schedule for a loan with loan principal 600,000 EUR, interest 4%, time to maturity 20 years as a
 - bullet loan
 - constant repayment loan
 - annuity loan

“Unscheduled”/Individual Repayment

- Individual repayment schedules
 - No repayment during general duration, repayment after termination of the loan (“balloon payment”)
 - Flexible repayments during duration of the loan agreement
 - Repayment via use of alternative financing sources
- Lender’s view on (unplanned) repayments
 - As the lender re-finances his investment, he might not be willing to accept flexible payments as he might not be able to repay his own obligation. He will then make a loss, as he needs to continue paying his finance cost to his lender.
 - The borrower therefore needs to pay an (early) (p)repayment penalty.

Flexible Interest

- Within a flexible interest loan the interest is fixed only for a short period, e. g. several months.
- The loan agreement itself can nevertheless be long-term.
- The interest is (re-)calculated at given points in time (e. g. every month, every quarter) and then applied to the outstanding loan balance.
- The calculation is based upon an agreed standard interest (frequently Euribor) plus a margin for the lender, e. g. Euribor + 50bps.
- Flexible interest contracts are frequently combined with flexible repayment schedules.

Financing with Bonds – Characteristics

- Debt financing with bonds is a financing form without taking a loan at one bank but rather issuing interest bearing papers to investors.
- A bond is therefore a security that typically offers an ongoing interest payment and includes the repayment of the principal amount after a given time period.
- A bond can be seen as a security that is similar to a share (e. g. regulation and supervision, multitude of investors) but without decision rights.
- A bond can be seen as a financing form that is similar to an amortisation free loan.
- In praxis many different forms of bonds can be used, e. g. zero-bonds (with no ongoing interest) or convertible bonds (where the bond buyer has additional right to convert the bond into shares).

Bonds – Valuation

- After being placed bonds are traded at capital markets.
- The market value of a bond is a function of the three factors that determine present value:
 - duration of the bond
 - interest rate of the bond (coupon)
 - market rate of interest
- If the market rate of interest is increasing (decreasing) the value of the bond is decreasing (increasing).
- Additional characteristics of a bond (e. g. convertible bonds, assets backed bonds) do also affect the market value of the bond.

Financing Structures Assessment

	Equity financing	Debt financing
Internal financing	+ no loss of influence - Problem for shareholders looking for dividends	+ relatively easy to implement - only temporary effects
External financing	+ increases credit rating - Change in voting rights possible - does not necessarily increase liquidity	+ no change in voting rights - Obligation to pay interest and repay principal amount

Cost of Equity

- The investor's expected rate of return shows his individual approach to an investment in the company (and not necessarily his judgement of the individual investment).
- This is also referred to as “cost of equity”, when the company uses equity finance.
- Additional to a risk free rate, risk is respected by using a measure for the individual company risk.
- Within this approach a risk-related “ β ” can be derived from the Capital Asset Pricing Model (CAPM).

Cost of Equity

The unsystematic risk of a company is measured by a company β which indicates the volatility of corporate value (and not real estate value or real estate volatility) in relation to the volatility of the whole market:

$$E(r_A) = r_f + [E(r_M) - r_f] * \beta_A$$

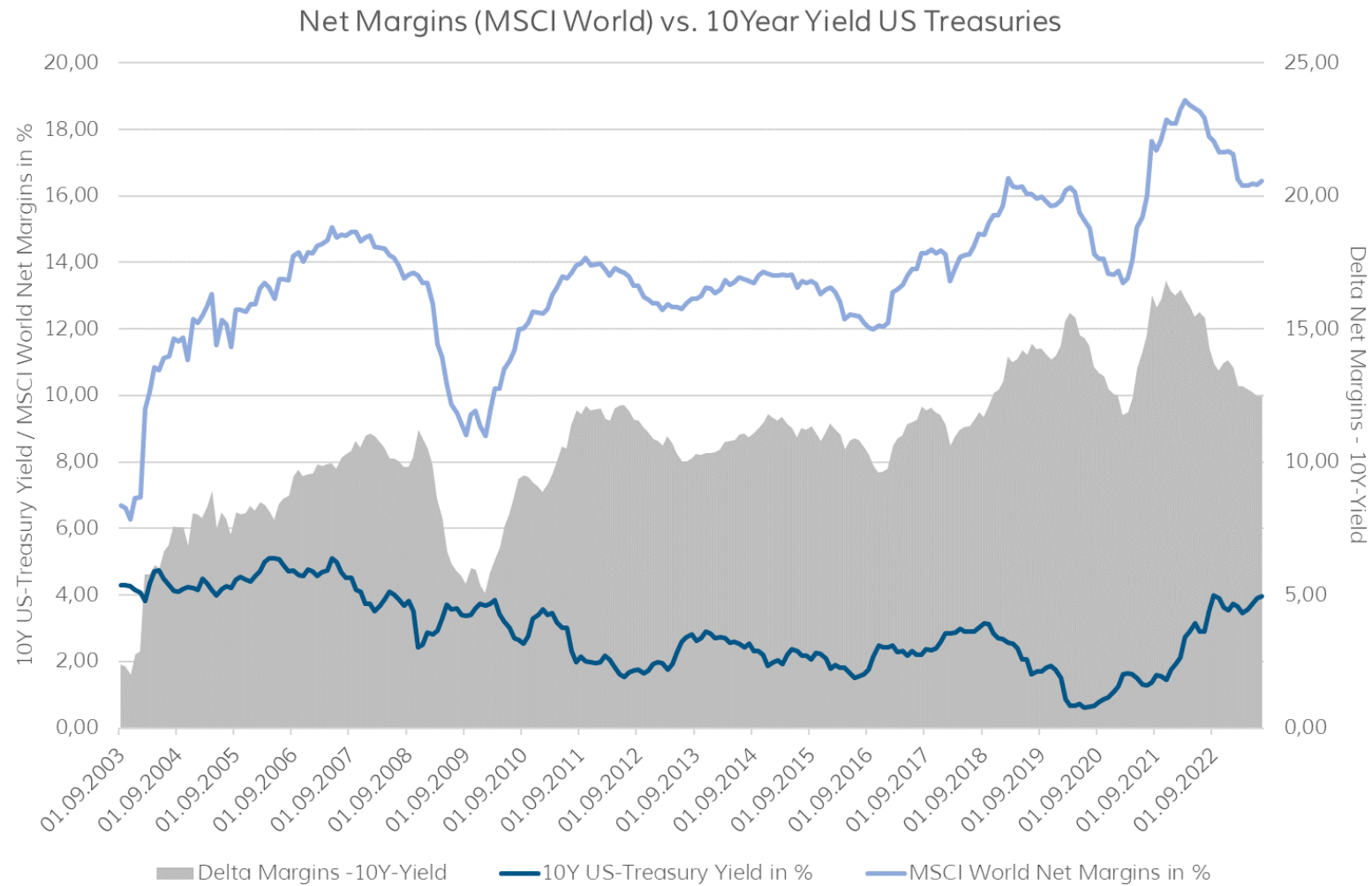
$E(r_A)$ = expected return on equity in company A (“cost of equity”)

r_f = risk free rate of return (“safe investment in state bonds”)

$E(r_M)$ = expected return on the market portfolio (“all companies”)

β_A = beta of company A (“risk of company A”)

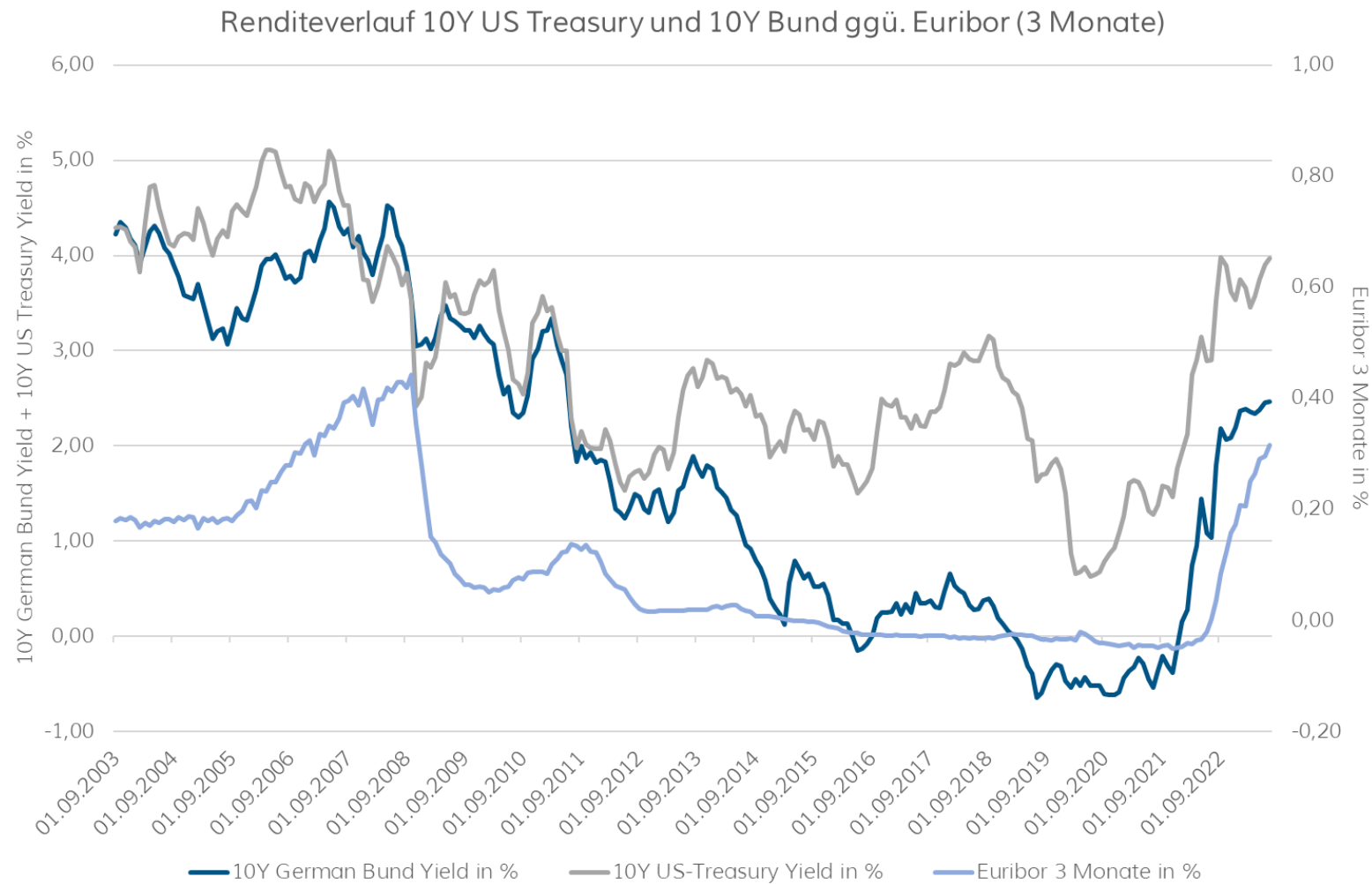
Development of Risk Premiums



Cost of Debt

- The borrower's expected rate of return shows his individual approach to an investment in the company. It consists of different elements:
 - Refinancing cost (= risk free rate)
 - Premium for risk (individual company risk for default, lower than equity related risk premiums)
 - Surcharge for administration
 - Margin for profit
- The total premium above the refinance cost is often referred to as “margin” and expressed in basis points (bps), where 1% = 100 bps.
- From the company's point of view, this is the “cost of debt”.

Development of Interest Rates



Cost of Total Capital

- If equity and debt are used in a company, the total cost of capital can be calculated based on the quotes of equity and debt in the financing.
- The total cost of capital are typically called weighted average cost of capital (wacc). It is computed by:

$$\text{wacc} = \frac{\text{EQ}}{\text{CE}} * \text{RoE} + \frac{\text{Debt}}{\text{CE}} * i \quad \text{OR} \quad \text{wacc} = \frac{\text{EQ}}{\text{CE}} * \text{RoE} + \frac{\text{Debt}}{\text{CE}} * i * (1 - t)$$

with

EQ: Equity

Debt: Debt

CE: (Total) Capital employed

RoE: Expected return on equity (= cost of equity)

i: Interest rate (= cost of debt)

t: tax quote

- Note that the second part is often multiplied with the tax quote t.

Cost of Total Capital – Example

- A company has 50 Equity and 150 Debt.
 - The expected return on equity is 10%, the interest rate is 5%.
 - The tax quote is 30%.
- ? Calculate the wacc without and with respect of taxes.

$$\text{wacc} = \frac{50}{50 + 150} * 10\% + \frac{150}{50 + 150} * 5\% = 6.25\%$$

$$\text{wacc} = \frac{50}{50 + 150} * 10\% + \frac{150}{50 + 150} * 5\% * (1 - 30\%) = 5.125\%$$

The Optimal Financing Structure

- Equity is more expensive than debt.
- With increasing leverage the bankruptcy risk of a company is increasing. This is the negative leverage effect, also known as leverage risk.
- As shareholders and creditors take this into account, the cost for equity and debt are increasing with increasing leverage.
- The question whether a company has an optimal financing structure (i. e. a leverage where financing cost is the lowest possible) is of special interest in the academic discussion.

Special Forms of Financing – Leasing

- Leasing is a contract between a lessor and a lessee to finance an investment the lessee is planning.
- The lessee pays a (monthly, annual) rent to the lessor instead of buying the investment asset directly.
- The cash and the income structure are changed:

	Acquisition	Leasing
Investment	Yes	No
Depreciation	Yes	No
Continuous rent payments	No	Yes
Capitalisation in balance sheet	Yes	No (partly yes)

Special Forms of Financing – Leasing

- Typical leasing forms are
 - Operating leasing with a short-term contract and rent payments that do not cover all expenses of the lessor.
 - Financing leasing with mid-term to long-term contracts and rent payments that can either cover all expenses of the lessor including his profit (full amortisation) or not (partial amortisation).
- Based on the concrete contract the leasing asset is to capitalise either on the lessor's or the lessee's balance sheet.

Leasing – Example

- A company wants to buy a machine for 100,000 €.
- Expected lifetime/leasing duration is 5 years.
- Interest for a bank loan is 5% p. a.
- The lessor calculates with 8% interest.
- Earnings before tax and investment related costs are 50,000 €.
- The quote for corporate tax and for business tax is 15%.
- ? Calculate the annual leasing fee.
- ? Which aspects need to be respected when comparing acquisition and leasing?
- ? Compare acquisition and leasing.

Leasing – Solution

- The annual leasing fee is the annuity of the initial payment over the expected lifetime/leasing duration.

$$L = 100,000 * [(1.08)^5 * 0.08] / [(1.08)^5 - 1] = 25,046 \text{ €}$$

- For the comparison between acquisition and leasing the following aspects are relevant:
 - Interest rate loan / Imputed costs for equity
 - Leasing fee / Calculation principles of the lessor
 - Addition interest / leasing payments for business tax
 - Depreciation and liquidation proceeds
 - Net present value of annual profits

Special Forms of Financing – Factoring

- Factoring means the sale of (existing or new) Receivables to third party (factoring company).
- The selling company receives liquidity (which leads to a qualification as financing instrument), the factoring company's earnings are based on its services.
- The factoring company can offer three different services:
 - Financing function (Finanzierungsfunktion)
 - Servicing function (Dienstleistungsfunktion)
 - Guarantee function (Delkrederefunktion)

Mezzanine Finance – General Description

- There is no legal or consistent definition for “mezzanine capital”.
- Mezzanine capital is a combination of equity and debt finance. It can also be called “hybrid capital”.
- Mezzanine financing is chosen, if lenders are not willing to give a higher amount of debt and/or equity is not available or its cost are too high.
- Mezzanine finance is basically debt capital that gives the lender the rights to convert to an ownership interest in the company if the loan is not paid back on time or completely.
- The typical debt (loan) is rather long-term, has a securitisation and a right to be repaid.
- The typical equity (share) is even longer-termed and has no right for repayment.

Mezzanine Finance – General Description

- Mezzanine capital is generally subordinated to debt and therefore positioned between equity and debt
 - either as junior / second-tier / subordinated debt
 - or as preferred debt / overriding equity
- The cost for mezzanine financing is higher than for typical debt and lower than for typical equity.
- Frequently a profit participation for the mezzanine tranche is agreed upon.
- Mezzanine financing can either be capital market oriented or on a private, individual basis.

Mezzanine Finance – Idealised Structure

Form of finance	LTV	Ca. Cost p.a.		Value
„Own“ equity tranche	rest	individual		
Private equity tranche	5-10%	15-20%		
Mezzanine tranche	5-15%	8-12%		
Junior debt	10-20%	5-8%		
Senior debt	ca. 60%	3-5%		

Mezzanine Finance – Economic Criteria

- Investment volume
- Margins, rates and fees
- Duration
- Amortisation
- Accounting treatment
- Taxation issues
- Loan-to-value (LTV) or Loan-to-cost (LTC)
- Interest coverage rate (ICR)
- Debt service coverage rate (DSCR)

Mezzanine Finance – Legal Criteria

- Treatment as loan or asset participation: Ideally, a mezzanine instrument
 - should be considered economically as equity,
 - but should be treated as debt by the tax authorities
- Participation in (variable) profits or ongoing (fixed) interest
- Participation in losses
- Decision rights

Mezzanine Finance – Legal Forms

- Subordinated loans (Nachrangdarlehen)
- Convertible bonds (Wandelanleihen)
- Option-linked bonds / Warrant-linked bonds (Optionsanleihen)
- Profit participation rights (Genussrechte, Genussscheine)
- Typical or untypical silent partnership (typisch oder atypisch stille Beteiligung)

Mezzanine Finance – Example

- A company plans an investment of 10 Mio. €. It gets the following offers:

Tranche	LTV	Margin
1	60%	50 basis points (bps)
2	61-80%	200 bps
3	81-90%	500 bps
4	91-100%	1000 bps

- ? The company is able to calculate the average interest cost for the tranches and in total.
- ? Calculate the average interest.

Mezzanine Finance – Solution

Tranche	Volume in €	Interest in %	Interest in €
1	6,000,000	$0.5\% + 0.5\% = 1\%$	60,000
2	2,000,000	$0.5\% + 2.0\% = 2.5\%$	50,000
3	1,000,000	$0.5\% + 5.0\% = 5.5\%$	55,000
<u>4</u>	<u>1,000,000</u>	<u>$0.5\% + 10.0\% = 10.5\%$</u>	<u>105,000</u>
Total	10,000,000	2.70%	270,000

Learning Objectives Chapter 5

- ✓ Structure and Explain different forms of financing
- ✓ Learn and Apply internal financing with equity and debt
- ✓ Learn and Apply external financing with equity
- ✓ Learn and Apply external financing with debt
- ✓ Understand combinations of financing forms and cost of capital
- ✓ See and Apply additional forms of financing

Agenda

1. Fundamentals on the Corporate and on Corporate Finance
2. Fundamentals on Financial Markets
3. Investment Valuation / Capital Budgeting
4. Calculation of Capital Needed
5. Corporate Financing
- 6. Corporate Analysis**
7. International Corporate Finance
8. Derivative Financial Instruments

Learning Objectives Chapter 6

- Recapitulate interdependencies between major corporate goals
- Get to know goals and process of corporate analysis
- Identify important items to analyse companies
- Develop profitability business ratios and Apply them to corporate examples
- Develop financial business ratios and Apply them to corporate examples

Content

6. Corporate Analysis

6.1 Fundamentals on Corporate Analysis

6.2 Profitability Analysis

6.3 Financial Analysis

Fundamentals on Corporate Analysis

- Corporate Analysis means the analysis of a company based on business ratios, combinations of business ratios and other measures.
- Goal is to analyse whether a company has been (ex post) or will be (ex ante) able to fulfil its planned economic (strategic, profitability, liquidity) goals.
- In analysing the three different goal dimensions potential, success and liquidity, three different levels of analyses can be derived:
 - Potential => Strategic analysis
 - Success => Profitability analysis
 - Liquidity => Financial analysis

Fundamentals on Corporate Analysis

- Applications of corporate analysis
 - Assessment of creditability
 - Identification of potentials threats on business continuity
 - Decisions on transactions (M&A)
 - Investment decisions
 - Use of business ratios as internal management system
- Addressees of corporate analysis
 - Management
 - (Potential) Shareholders
 - Employees
 - Analysts
 - Creditors, banks

Process of Corporate Analysis

1. Setting of analysis goals
2. Preparation of information
 - Elimination of accounting policy measures
 - Elimination of one-off effects
 - Preparation according to timeliness
 - Results will be structured balance sheets and cash flow statements
3. Execution of corporate analysis
4. Interpretation of results

Profitability Analysis

- The aim is to assess the earnings power of a company.
- Typically this is done based on corporate planning information.
- Alternatively existing financial statements are used.
- Analysed are
 1. Sources of profits
 2. Structure of profits
 3. Profitability analysis

Sources of Profits

- Economic split of profits

	Operational components	Non-operational components
Regular components	Operating result	Financial result
Irregular components	Extraordinary result	

- Operating result
- Financial result
- Extraordinary result
- Taxes

Profitability Analysis – Result levels

Very often companies do not only calculate and disclose the net profit but results on „higher“ levels to show operating results and to compare the performance to other companies.

Result	Interpretation	Example
Turnover	Market performance	1,100
- <u>Expenses</u>		<u>700</u>
= EBITDA	Operating result before depreciation	400
- <u>Depreciation</u>		- <u>50</u>
= EBIT	Operating result	350
- <u>Interest</u>		- <u>50</u>
= Profit before taxes		300
- <u>Taxes</u>		- <u>100</u>
= Net profit / Net loss	Effective result	200

Sources of Profits

- Result structure
 - Quote ord. Operating result = $\frac{\text{Operating result}}{\text{Total result before taxes}}$
 - Quote Financial result = $\frac{\text{Financial result}}{\text{Total result before taxes}}$
 - Quote extraord. Operating result = $\frac{\text{Extraordinary Operating result}}{\text{Total result before taxes}}$
- Analysis of expenses
 - Material intensity = $\frac{\text{Material expenses}}{\text{Revenues}}$
 - HR intensity = $\frac{\text{HR expenses}}{\text{Revenues}}$
 - Capital intensity = $\frac{\text{Depreciation}}{\text{Revenues}}$
 - Cost of debt = $\frac{\text{Interest expenses}}{\text{Interest bearing liabilities}}$
 - Tax quote = $\frac{\text{Tax expenses}}{\text{Result before taxes}}$

Profitability Analysis

- Revenue profitability = $\frac{\text{Net profit}}{\text{Revenues}}$
- Equity profitability (ROE) = $\frac{\text{Net profit}}{\text{Equity}}$
- Equity profitability (ROE) = $\frac{\text{Net profit}}{\text{Total capital}}$ (seldom)
- Return on capital employed (ROCE) = $\frac{\text{Net profit} + \text{Interest}}{\text{Total capital}}$
- Leverage index = $\frac{\text{EQ} - \text{profitability}}{\text{TC} - \text{profitability}}$

Profitability Analysis – Example

Balance sheet 01.01.			
Fixed assets	500,000	Share capital	100,000
Inventories	500,000	Paid-in capital	900,000
Acc. receivables	500,000	Long-term debt	500,000
Bank	500,000	Short-term debt	500,000

Income statement 01.01.-31.12.			
Material expenses	400,000	Revenues	1,000,000
HR expenses	300,000	Other earnings	100,000
Interest expenses	50,000		
Depreciation	50,000		
Taxes	100,000		

Balance sheet 31.12.			
Fixed assets	600,000	Share capital	100,000
Inventories	600,000	Paid-in capital	900,000
Acc. receivables	200,000	(Net/Balance) Profit	200,000
Bank	600,000	Long-term debt	300,000
		Short-term debt	500,000

Profitability Analysis – Example

- Revenue profitability = $\frac{\text{Net profit}}{\text{Revenues}} = \frac{200,000}{1,000,000} = 20\%$
- Material intensity = $\frac{\text{Material expenses}}{\text{Revenues}} = \frac{400,000}{1,000,000} = 40\%$
- HR intensity = $\frac{\text{HR expenses}}{\text{Revenues}} = \frac{300,000}{1,000,000} = 30\%$
- Return on equity = Equity profitability = $\frac{\text{Net profit}}{\text{Equity}} = \frac{200,000}{1,000,000} = 20\%$
- Cost for debt = $\frac{\text{Interest}}{\text{Bankdarlehen}} = \frac{50,000}{1,000,000} = 5\%$
- Return on capital employed (ROCE) = $\frac{200,000+50,000}{2,000,000} = 12.50\%$

Financial Analysis

- The aim is to assess the liquidity of a company.
- The liquidity is the availability of a company to meet its financial obligations at any time.
- Levels of the analysis are
 1. Investment analysis: Analysed is the structure of assets as well as the lockup time of capital. (“When is the asset converted into cash?”)
 2. Finance analysis: Analysed is the structure of capital as well as the cost of capital. (“When must the obligation be paid?”)
 3. Liquidity analysis: Analysed is the timing relation between assets and liabilities. (“Is cash available to pay obligations?”)

Investment Analysis

- Current assets – intensity = $\frac{\text{Current assets}}{\text{Total assets}}$
- Non – current assets – intensity = $\frac{\text{Fixed assets}}{\text{Total assets}}$
- Asset structure = $\frac{\text{Fixed assets}}{\text{Current assets}}$
- Inventories lock up = $\frac{\text{Inventories}}{\text{Revenues}}$
- PPE lock up = $\frac{\text{PPE assets}}{\text{Revenues}}$
- Receivables lock up = $\frac{\text{Acc.receivables}}{\text{Revenues}}$
- Turnover frequency Inventories = $\frac{\text{Material expenses}}{\text{Average stock}}$
- Turnover time Inventories = $\frac{\text{Average stock}}{\text{Material expenses}} * 365$

Finance Analysis

- Equity – Quote = $\frac{\text{Equity}}{\text{Total capital}}$
- Debt – Quote = $\frac{\text{Debt}}{\text{Total capital}}$
- Leverage = $\frac{\text{Debt}}{\text{Equity}}$

Liquidity Analysis based on Balance Sheet

Golden rule of balance sheet :

$$\text{Longterm assets} \leq \text{Equity} + \text{longterm debt}$$

Long-term liquidity

- Cover ratio A = $\frac{\text{Equity}}{\text{Fixed assets}}$
- Cover ratio B = $\frac{\text{Equity} + \text{Long-term debt}}{\text{Fixed assets}}$

Short-term liquidity

- Liquidity 1 = Cash ratio = $\frac{\text{Bank (= cash assets+cash equivalents)}}{\text{Short-term debt}}$
- Liquidity 2 = Quick ratio = $\frac{\text{Bank+Acc. receivables}}{\text{Short-term debt}}$
- Liquidity 3 = Current ratio = $\frac{\text{Current assets}}{\text{Short-term debt}}$
- Net working capital: Current assets – Shortterm debt

Liquidity Analysis based on Cash Flows

Net profit

+ Depreciation

+/- Increase/Decrease Provisions

-/+ Increase/Decrease Current assets (= the so called „Working Capital“: Inventories and Acc. receivables)

-/+ Increase/Decrease Fixed assets

+/- Increase/Decrease Debt

+/- Increase/Decrease Equity

= Cash Flow

Cash Flow Analysis

- Cash Flow profitability = $\frac{\text{Cash Flow}}{\text{Total capital}}$
- Cash Flow – Turnover rate = $\frac{\text{Cash Flow}}{\text{Revenues}}$
- Dividend power = $\frac{\text{Dividends}}{\text{Cash Flow}}$
- Debt repayment potential = $\frac{\text{Cash Flow}}{\text{Debt}}$
- Interest Service Coverage Rate (ISCR) = $\frac{\text{Cash Flow}}{\text{Interest}}$
- Debt Service Coverage Rate (DSCR) = $\frac{\text{Cash Flow}}{(\text{Interest} + \text{Repayments})}$

Financial Analysis – Example

Balance sheet 01.01.			
Fixed assets	500,000	Share capital	100,000
Inventories	500,000	Paid-in capital	900,000
Acc. receivables	500,000	Long-term debt	500,000
Bank	500,000	Short-term debt	500,000
Income statement 01.01.-31.12.			
Material expenses	400,000	Revenues	1,000,000
HR expenses	300,000	Other earnings	100,000
Interest	50,000		
Depreciation	50,000		
Taxes	100,000		
Balance sheet 31.12.			
Fixed assets	600,000	Share capital	100,000
Inventories	600,000	Paid-in capital	900,000
Acc. receivables	200,000	(Net/Balance) Profit	200,000
Bank	600,000	Long-term debt	300,000
		Short-term debt	500,000

Financial Analysis – Example

Investment analysis

- Current assets – intensity = $\frac{\text{Current assets}}{\text{Total assets}} = \frac{1,500,000}{2,000,000} = 75\%$
- Turnover frequency Inventories = $\frac{\text{Material expenses}}{\text{Average stock}} = \frac{400,000}{(500,000+600,000)/2} = 0,72 \text{ (times)}$
- Turnover time Inventories = $\frac{\text{Average stock}}{\text{Material expenses}} * 365 = \frac{(500,000+600,000)/2}{400,000} * 365 = 501 \text{ (days)}$

Financial analysis (as of 01.01.)

- Equity – Quote = $\frac{\text{Equity}}{\text{Total capital}} = \frac{1,000,000}{2,000,000} = 50\%$
- Leverage = $\frac{\text{Debt}}{\text{Equity}} = \frac{1,000,000}{1,000,000} = 1$

Financial Analysis – Example

Cash Flow Analysis

- Cash Flow 01.01. – 31.12. =?

Cash Flow 01.01.-31.12.	
Net profit	+ 200,000
+ Depreciation	+ 50,000
- Increase Non-current assets	- 150,000
- Increase Inventories	- 100,000
+ Decrease Acc. Receivables	+ 300,000
<u>- Decrease Long-term debt</u>	<u>- 200,000</u>
= Cash Flow	+ 100,000

Learning Objectives Chapter 6

- ✓ Recapitulate interdependencies between major corporate goals
- ✓ Get to know goals and process of corporate analysis
- ✓ Identify important items to analyse companies
- ✓ Develop profitability business ratios and Apply them to corporate examples
- ✓ Develop financial business ratios and Apply them to corporate examples

Agenda

1. Fundamentals on the Corporate and on Corporate Finance
2. Fundamentals on Financial Markets
3. Investment Valuation / Capital Budgeting
4. Calculation of Capital Needed
5. Corporate Financing
6. Corporate Analysis
- 7. International Corporate Finance**
8. Derivative Financial Instruments

Learning Objectives Chapter 7

- Get to know historical development of exchange rate systems
- Understand macroeconomic influences on exchange rates
- Analyse exchange rates interdependencies
- Transfer general thoughts on investment appraisals on specific international aspects
- Adopt corporate financing on the needs of internationally active companies
- Get to know risk management measures for internationalisation

Content

7. International Corporate Finance

7.1 Currency Systems and Exchange Rates

7.2 International Aspects of Investment Valuation

7.3 International Aspects of Corporate Financing

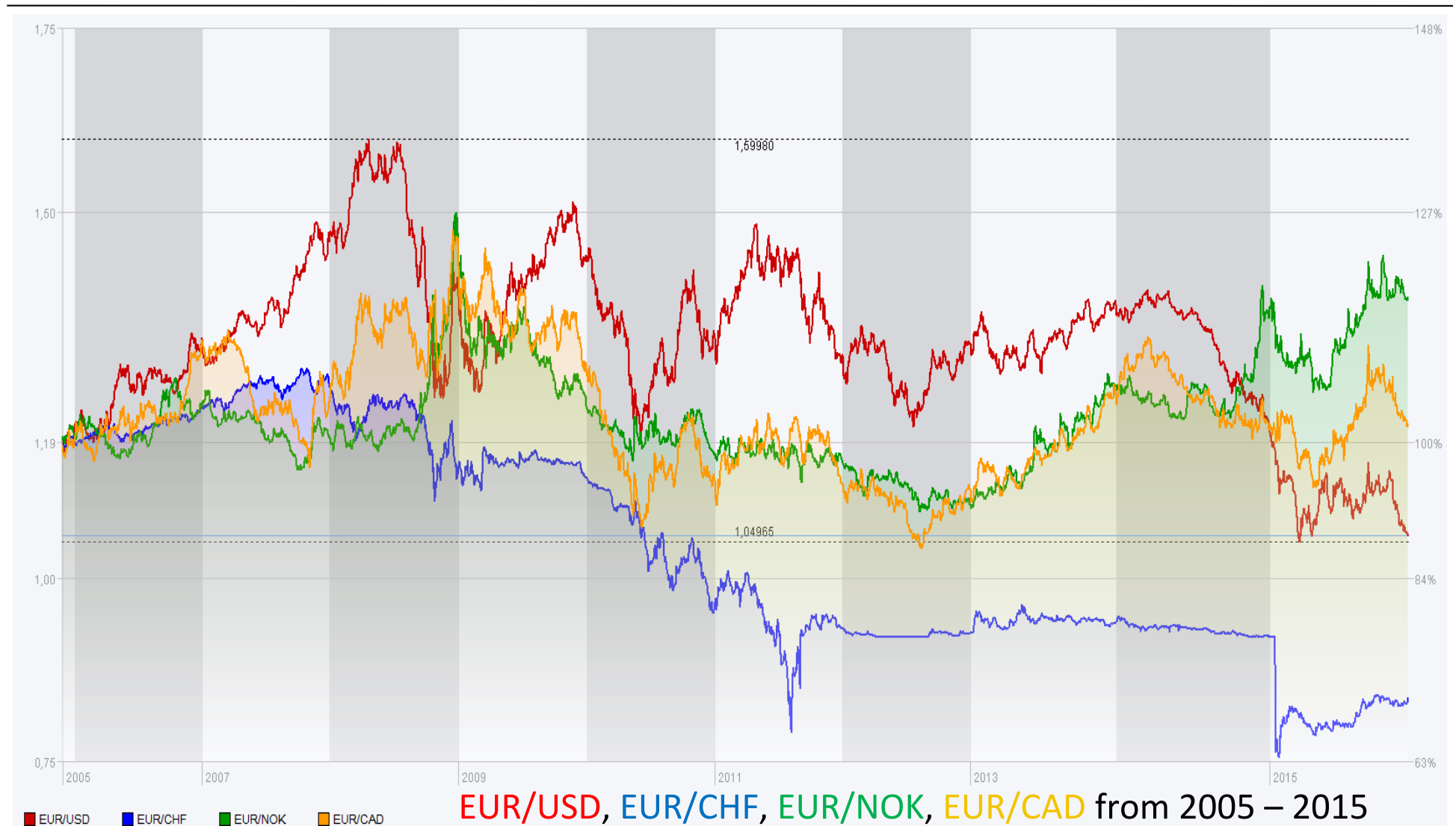
Exchange Rate Systems

- Gold standard: Before and after WW I most currencies were backed by gold, this means money could be exchanged into gold at any time, respectively money could only be issued, if gold reserves were existing.
- Bretton Woods was an international treaty after WW II that was the fundament of an exchange rate system with a fixed exchange rate of the single currencies to the US-\$ and a fixed conversion rate from the US-\$ to the price of gold (1 ounce = 35 US-\$).
- After its collapse in 1973 most exchange rates were made flexible.
- Today most currencies can be exchanged without restrictions.

Exchange Rates

- The exchange of two (convertible) currencies is executed with the exchange rate.
- The exchange rate is influenced by different, mostly macroeconomic factors:
 - Purchase power / Inflation rate
 - Interest rates
 - Countries' balance of activities (trade deficits)
 - Expectations of market participants
- If a currency loses value, it is “depreciated”, value gains lead to an “appreciation”.
- Always respect the notation of exchange rates:
 - Indirect quotation: Amount of foreign currency for 1 Euro, e. g. EUR/USD: 1 € = 1.075 US-\$ or EUR/GBP: 1 € = 0.71 GBP
 - Direct quotation: Price for foreign currency in 1 Euro, e. g. USD/EUR: 1 US-\$ = 0.93 € or GBP/EUR: 1 GBP = 1.40 €
- Another differentiation is bid rate (bank sells foreign currencies) and ask rate (bank buys foreign currencies)

Development of Exchange Rates



Link Spot Rate and Future Rate

- The spot rate indicates the exchange rate today, while the future rate shows exchange rate at a given point in future.
- The link can be demonstrated with the (covered) interest parity theory:
 - If interest rates in two countries are different, money will be invested in the currency offering a higher return.
 - As both currencies must –in theory– have the same “internal value” (i. e. purchase power), the currency with the higher interest rate loses value due to inflation.
 - Therefore it will be depreciated compared to the other currency.
 - This will lead to an adjustment of exchange rates (in the mid-term).
- These thoughts lead to an arbitrage free prognosis of the future exchange rate:

$$\frac{\text{EUR}}{\text{USD}}_{\text{Future}} = \frac{1 + i_{\text{USD}}}{1 + i_{\text{EUR}}} * \frac{\text{EUR}}{\text{USD}}_{\text{Spot}}$$

Link Spot Rate and Future Rate

- Example: On 1.1. the exchange rate EUR/USD is 1.10. US bonds offer a return of 3%, German bonds 1%.
- ? What will be the exchange rate on 31.12.?
- The exchange rate will change due to the difference in interests for $\frac{1+0.03}{1+0.01} = 1.98\%$ to EUR/USD 1.122.
- ? What would you do, if you were offered a future exchange rate of (I.) EUR/USD 1.11 or (II.) EUR/USD 1.13?
- I. You take a loan of 100 € @ 1%, exchange this amount into 110 US-\$ and invest this @ 3%. On 31.12. you exchange the 113.30 US-\$ for EUR/USD 1.11 into 102.70 € and repay the loan with 101.00 €.
- II. You take a loan of 110 US-\$ @ 3%, exchange this amount into 100 € and invest this @ 1%. On 31.12. you exchange the 110 € for EUR/USD 1.13 into 114.13 US.\$ and repay the loan with 113.30 US-\$.

Aspects of Internationalisation

- Political aspects
 - Macro-political risks
 - Micro-political risks
- Legal, financial and tax aspects
 - Legal forms
 - Dividend possibilities
 - Taxation / Double tax treaties
 - Exchange rates / Financial reporting
- Structural and operative aspects
 - Organisation of international activities
 - Risk management of international activities
 - Cross cultural management

Risks of Internationalisation

- Business risk
 - Default of business partner
- Political risk
 - Macro risks
 - Default of the state
 - Convertibility of currency
 - Transferability of money
 - Cultural and institutional risk
 - Micro risks: company or sector-specific
- Exchange rate risks
 - (short-term) Transaction risk
 - (mid-term) Operations risk
 - Balance sheet risk

International Investment Valuation

- The cash flows out of international activities are subject to “regular” project risks and additionally to a currency risk.
- When using the NPV method for investment decisions, two alternative procedures can be applied
 1. Calculate NPV in the foreign currency with the foreign cost of capital (“local wacc”) and convert the result into the home currency based on the current spot rate.
 2. Convert future results into the home currency based on future rates and calculate the NPV with the home cost of capital (“home wacc”).
- Additionally timing issues and tax issues need to be respected.

International Corporate Financing

- Financing in foreign currencies
- Placement of (long-term) bonds on international capital markets
- Implementation of “non-standard” financing forms
 - Convertible bonds
 - Structured finance / Asset backed securities
- Listing in the U.S.
- Cross border Working Capital Management and Cash Management based on the (expected) development of the exchange rates

Foreign Currency Loan – Example

- A company is offered two different loans of 1 Mio. €. The interest rate for a German loan is 3%. The interest rate for a comparable loan in Swiss francs is 1%. The repayment is after 5 years. The exchange rate EUR/CHF is 1.20.
 - ? Calculate the amount needed for the Swiss loan.
 - ? What is the interest advantage of the Swiss loan within the five years (in €)?
 - ? What is the maximum change in the exchange rate so that the finance in Swiss francs is still more profitable?

Foreign Currency Loan – Example

- $\text{EUR/CHF} = 1,20 \Rightarrow 1 \text{ EUR} = 1,20 \text{ CHF} \Rightarrow 1,000,000 \text{ EUR} = 1,200,000 \text{ CHF}$
- Interest EUR loan: $10 * 1,000,000 \text{ €} * 3\% = 30,000 \text{ EUR}$
Interest CHF loan: $10 * 1,200,000 \text{ €} * 1\% = 12,000 \text{ CHF} \Rightarrow 12,000 \text{ CHF} = 10,000 \text{ EUR}$
The interest advantage is $(30,000 \text{ €} - 10,000 \text{ €}) = 20,000 \text{ EUR p. a.}$
- The repayment must not increase by more than $5 * 20,000 \text{ EUR} = 100,000 \text{ EUR}$, i. e. a maximum repayment of 1,100,000 EUR is allowable.

$\text{CHF loan} = 1,200,000 \text{ CHF} = 1,100,000 \text{ EUR} \Rightarrow \text{EUR/CHF} = 1.09.$

If the CHF is appreciating more (EUR/CHF goes below 1.09), more than 1,100,000 EUR must be paid back and the foreign currency loan is not profitable.

Internationalisation – Risk Management

- Funding and investing in different currencies
- Implementation of financial derivatives
 - Forwards
 - Futures
 - Options
 - Currency swaps
 - Interest rate swaps
- Use of hybrid financial instruments

Learning Objectives Chapter 7

- ✓ Get to know historical development of exchange rate systems
- ✓ Understand macroeconomic influences on exchange rates
- ✓ Analyse exchange rates interdependencies
- ✓ Transfer general thoughts on investment appraisals on specific international aspects
- ✓ Adopt corporate financing on the needs of internationally active companies
- ✓ Get to know risk management measures for internationalisation

Agenda

1. Fundamentals on the Corporate and on Corporate Finance
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8. **Derivative Financial Instruments**

Learning Objectives Chapter 8

- Learn definition and terms of financial derivatives
- Get to know participants in the market for derivatives and understand reasons for their application
- Understand basic functionality of different derivative products
- Analyse the profit and loss schedules of different derivative products
- Discuss differences between different derivative products
- Apply derivative functionalities on basic examples

Content

8. Derivative Financial Instruments

8.1 Definition of and Reasons for the Use of Financial Instruments

8.2 Characteristics of selected Derivative Financial Instruments

Derivatives – Terms and Definitions

- Derivatives are financial instruments that directly “derive” their value from changes in the value of a related asset or liability, called the underlying.
- The price of the derivative changes when the price of the underlying changes.
- Because derivatives may consist of several types of financial assets, changes in the price of the derivative may not be proportional to changes in the values of the underlying assets.

Derivatives – Terms and Definitions

- Long position is the buyer's position.
- Short position is the seller's (or "writer's") position.
- Underlying is an asset / a variable (prices, rates, ...) on which the price of a derivative instrument depends.
- Notional amount is the number of units or quantity of underlying that are specified in the derivative instrument. It is necessary to calculate the value of the derivative instrument.
- Exercise price (or strike price) is the price agreed (at the beginning) at which a transaction would be made.

Derivatives – Terms and Definitions

- Trade date is the date in which parties agreed the contract.
- Effective date is the starting date of the contract's effects.
- Termination date is the maturity date of the contract.
- Payment (or settlement) date is the date in which there is a cash flow exchange between parties related to the result of the derivatives.

Derivatives – Reasons

- Hedging
 - Reduction of risk / transfer of a risk to another party
 - A derivative works like a “warranty” on the future price of the underlying.
 - The price for hedging is either a reduction of chance (“loss of potential profit”) or a direct payment of a premium (“insurance premium”).
- Speculation
 - Speculators show speculative behaviour and take risk to generate high(er) return.
 - The profit depends on the fulfilment of subjective market expectations.
 - Speculation may result in substantial loss.

Derivatives – Reasons

- Arbitrage
 - Arbitraders take advantage of price differences between different markets.
 - This can be either regionally (Frankfurt/New York) or logically (spot market/forward market) to generate riskless profits.
- Portfolio diversification
 - The use of derivatives (like any other asset) to create a new combination within a portfolio can lead to further diversification.

Derivatives – Types

- Based on the nature of the underlying
 - Financial derivatives, in which the underlying is an intangible asset (stocks, interest rates, indexes, currencies...)
 - Commodity derivatives, in which the underlying is a tangible asset (gold, steel, coffee, wheat, oil...)
 - Weather derivatives in which the underlying can be rain, temperature
- Related to the place of trading
 - quoted in regulated markets: exchange traded derivatives
 - Standardised contracts
 - Counterparts can be individuals or a financial institutions
 - Settlement systems and clearing houses are needed
 - Example: <http://www.eurexchange.com/>
 - traded over-the-counter (OTC): over-the-counter derivatives
 - Not standardised contracts
 - Counterparts are usually financial institutions

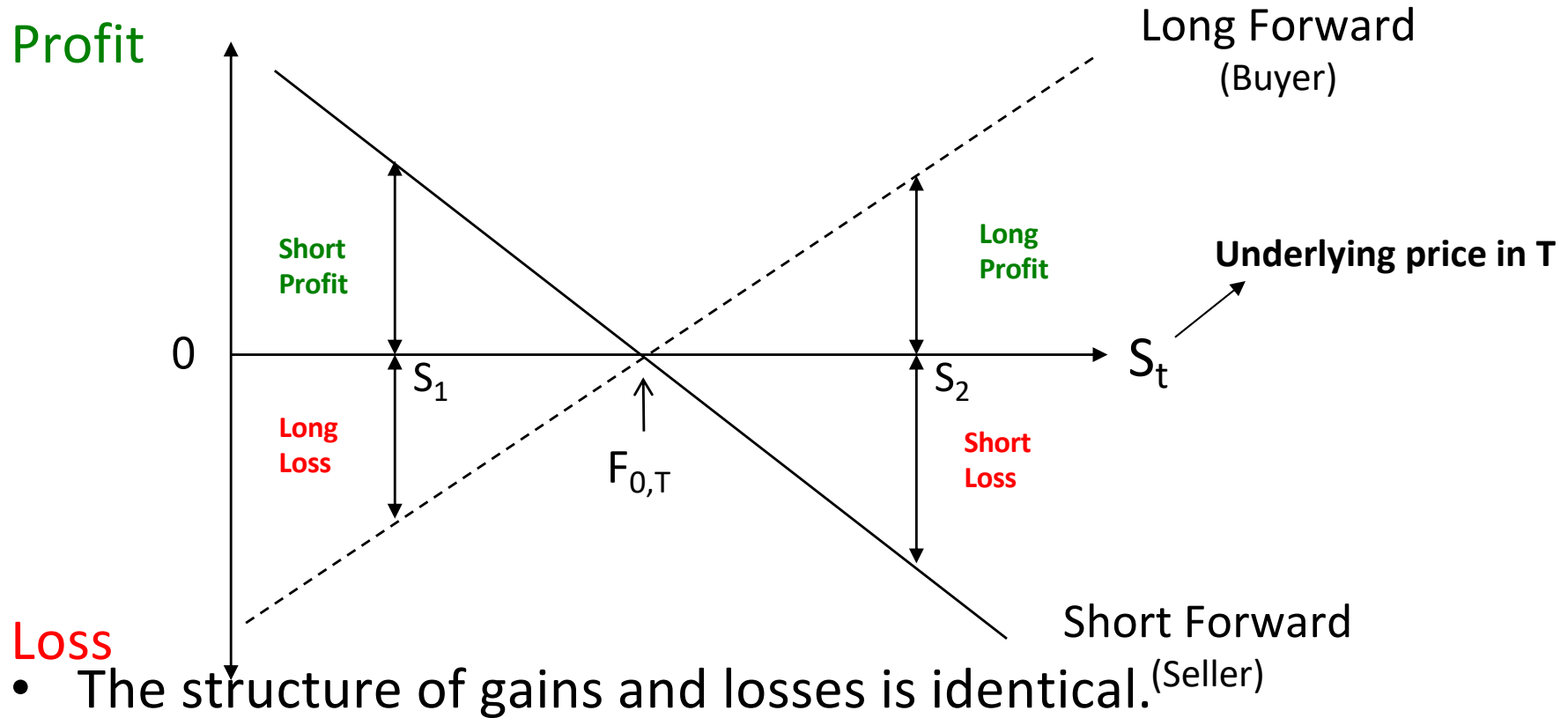
Forwards – Definition

- A forward contract is an agreement today (t_0) between two parties to exchange a specified amount of a commodity, security, or foreign exchange at a specified date in the future (t_T) with the forward price ($F_{0,T}$) being set now and the future price / spot price (S_T) unknown.
- In other terms, these contracts enable to buy or sell a specific quantity of a particular commodity, foreign currency or other financial instrument at its forward price now, with delivery and settlement at a specified future date at the future price.

Forwards – General Description

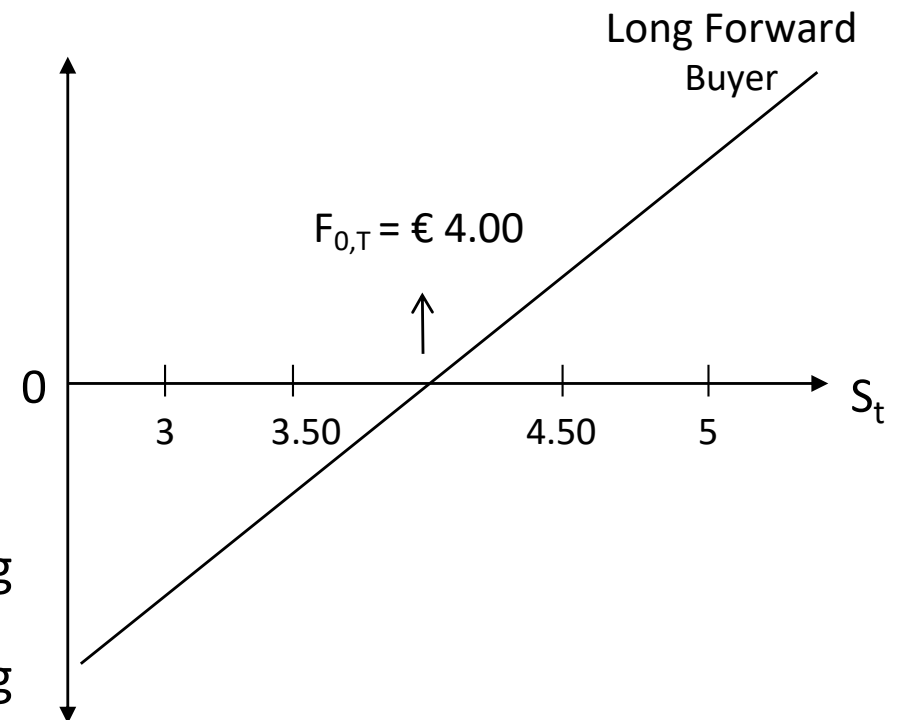
- Forwards are popular in currency exchange markets, but also used in commodity or financial markets.
- Forward contracts are traded over-the-counter, the terms are negotiable.
- Finding another counterpart is complicated, the market is therefore not liquid.
- Forwards are subject to credit risk or default risk.
- They imply one specific delivery date, which is the settlement at the end of the contract.
- A forward contract is symmetric (zero sum game): The profit of investor 1 (e. g. the buyer) is the loss of investor 2 (e. g. the seller).

Forwards – Figure



Forwards – Example

- The forward price for a commodity ($F_{0,T}$) is 4.00 €.
- A forward for 2,000 units is closed.
- The holder of the long position will gain the difference between the underlying spot price in t_T (S_T) and the forward price in t_0 ($F_{0,T}$).
- In T the holder of the long position must buy the 2,000 units from the holder of the short position at € 4,00.
 - If $S_T = 5.00$ €: The holder of the long position makes a profit of 2,000 €.
 - If $S_T = 3.00$ €: The holder of the long position makes a loss of 2,000 €.



Futures – Definition

- A future is an exchange traded contract which includes the obligation for one party to buy or sell a specified quantity of a commodity or a financial security at a specified price on a specified future date.
- In other terms, these contracts represent an agreement to buy or sell specific and standardised quantities of commodities or financial instruments at particular prices on stipulated future dates.

Futures – General Description

- Futures are unconditional contracts, that are standardised and traded on exchange markets.
- They are liquid: It is possible to “reverse” a position and offset the future obligation in every moment during the contract life.
- They are usually closed prior to maturity because investors are not interested in owning the underlying.
- Every underlying price changes involve changes in the contract payoff through a daily “marking to market” process.
- Futures do have less credit risk than other derivatives because initial margins and additional margins are paid.
- They do not have a settlement risk.
- Futures contracts are available both on commodities and financial assets: agricultural products, metals and oil, interest rates, currencies, stock market indices, etc.

Futures – Marking to Market

- The “marking to market” is the daily cash realisation of gains or losses in the value of the futures contract during the day.
- Within the marking to market process there is settlement after every trading date: The contract is rewritten at the new price, which reduces default risk.
- In order to settle the contract every single day, both parties need to pay a deposit to the clearing house:
 - An initial margin (fixed percentage of contractual nominal amount usually between 1% and 10%)
 - A maintenance (call) margin that varies daily depending on the changes in underlying price. If the maintenance margin goes below the minimum margin, it has to be restored by a new deposit (variation margin).

Futures – Example

- Consider a future contract where A buys from B 1,000 units of silver, within three days, at a fixed price of 6.00 € per unit.

The price of silver changes:

- 6.10 € on day 1
- 6.05 € on day 2
- 6.12 € on day 3

- ? How does the value of the future change?
- ? What payments are necessary to settle the contract?

Futures – Solution

- Day 1
 - B must pay to A $(6.10 \text{ €} - 6.00 \text{ €}) * 1,000 = 0.10 \text{ €} * 1,000 = 100 \text{ €}$ as the settlement for the first day's price increase of 0.10 €.
 - Once this payment is made, the contract is rewritten with a price of 6.10 €.
- Day 2
 - A must pay to B $(6.10 \text{ €} - 6.05 \text{ €}) * 1,000 = 0.05 \text{ €} * 1,000 = 50 \text{ €}$ as settlement for the second day's price decrease.
 - The contract is now rewritten at 6.05 €.
- Day 3
 - B must pay to A $(6.12 \text{ €} - 6.05 \text{ €}) * 1,000 = 0.07 \text{ €} * 1,000 = 70 \text{ €}$ as the settlement for the third day's price increase of 0.07 €.
- Summary
 - B has paid 100 € and 70 € and has received 50 € => the total net payment made by B is 120 €.
 - This equals to $(6.12 \text{ €} - 6.00 \text{ €}) * 1,000 = 0.12 \text{ €} * 1,000 = 120 \text{ €}$.
 - These settlements are a compensation for the profits/losses that occur due to the non-delivery of the underlying.

Forwards and Futures – Similarities

- Forward and Futures are both contracts with the obligation
 - to buy or sell a specific commodity or security
 - at a specific price
 - on a specific delivery date
- The buyer of these contracts must buy the commodity at the stated price and time (which means he is in the “long” position).
- The seller of these contracts must sell the commodity at the stated price and time (which means he is in the “short” position).

Forwards versus Futures – Differences

- Forwards
 - Not standardised
 - Traded on over-the-counter market
 - One specific delivery date
 - Settled at the end of the contract
 - Delivery or final cash settlement usually takes place
- Futures
 - Standardised contracts
 - Traded on an exchange market
 - Range of delivery dates
 - Usually the contract is closed out prior to maturity
 - Daily settled (“marking to market”): its value is always equal to zero

Options – Definition

- Options offer the buyer the right without obligation to buy or sell a financial instrument or a commodity at a fixed price before or on a specific date.
- A call option is a contract that gives the buyer the right to buy a security or a commodity (the underlying) at a stated exercise price on or before a stated expiration date.
- A put option is a contract that gives the buyer the right to sell a security or commodity (the underlying) at stated exercise price on or before a stated expiration date.

Options – General Description

- The buyer of a (buy or sell) option has the long position in the contract.
- The seller of a (buy or sell) option has the short position.
- The option premium is paid by the buyer to the seller for the option right (i. e. the right to buy or to sell).
- The buyer is the one to decide whether he uses his right or not, the seller has no decision right.
- The exercise price (or strike price) is the price agreed for buying or selling the underlying asset.
- The expiration date is the time limit of the option contract.
 - European options can be exercised only at expiration date.
 - American options can be exercised any time before expiration.

Options – Example Call Option

- The price of one share today is X € (e. g. 10 €).
- A call option C gives you the right to buy one share at the price of 12 € in one year.
- ? What expectations do you have if you buy the option (i. e. you are “long”) and get the right to buy the share?
- ? What expectations do you have if you sell the option (i. e. you are “short”) and maybe must sell the share?
- ? How does the profit/loss structure (and the option value) change with changes in the underlying share price?

Share price	8 €	10 €	12 €	15 €	100 €
Buyer/Long	-2 €	-2 €	-2 €	-2 € + 3 € = +1 €	-2 € + 88 € = +86 €
Seller/Short	+2 €	+2 €	+2 €	+2 € - 3 € = -1 €	+2 € - 88 € = -86 €

Options – Example Put Option

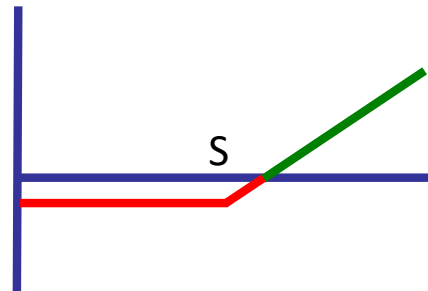
- The price of one share today is X € (e. g. 10 €).
- A put option P gives you the right to sell one share at the price of 12 € in one year.
- ? What expectations do you have if you buy the option (i. e. you are “long”) and have the right to sell the share?
- ? What expectations do you have if you sell the option (i. e. you are “short”) and maybe must buy the share?
- ? How does the profit/loss structure (and the option value) change with changes in the underlying share price?

Share price	0 €	8 €	10 €	12 €	15 €
Buyer/Long	$-2 \text{ €} + 12 \text{ €} = +10 \text{ €}$	$-2 \text{ €} + 4 \text{ €} = +2 \text{ €}$	$-2 \text{ €} + 2 \text{ €} = 0 \text{ €}$	-2 €	-2 €
Seller/Short	$+2 \text{ €} - 12 \text{ €} = -10 \text{ €}$	$+2 \text{ €} - 4 \text{ €} = -2 \text{ €}$	$+2 \text{ €} - 2 \text{ €} = 0 \text{ €}$	+2 €	+2 €

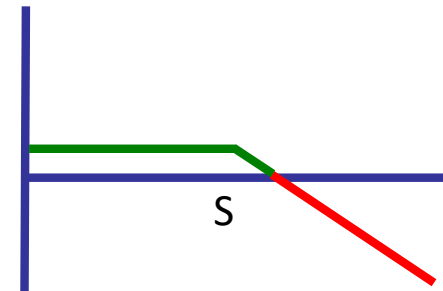
Options – Figures

- Call option

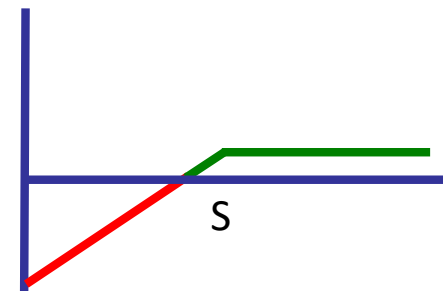
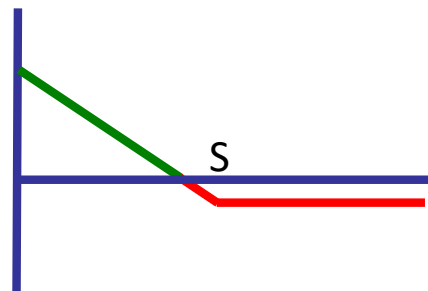
Buyer



Seller



- Put option



— Profit

— Loss

Options – Summary of Combinations

		Position in the contract	
		Buyer ("Long")	Seller ("Short")
Type of option	Call Option ("Buy underlying")	- The buyer expects price increases. - He pays the option premium to secure a "cheap" acquisition price. - His maximum loss is the option premium. - His maximum profit is unlimited.	- The seller expects price decreases. - He receives the option premium to make a profit. - His maximum profit is the option premium. - His maximum loss is unlimited.
	Put Option ("Sell underlying")	- The buyer expects price decreases. - He pays the option premium to secure a "cheap" sales price (or to secure existing shares against a loss in value). - His maximum loss is the option premium. - His maximum profit is the exercise price less the option premium.	- The seller expects price increases. - He receives the option premium to make a profit. - His maximum profit is the option premium. - His maximum loss is the exercise price less the option premium.

Learning Objectives Chapter 8

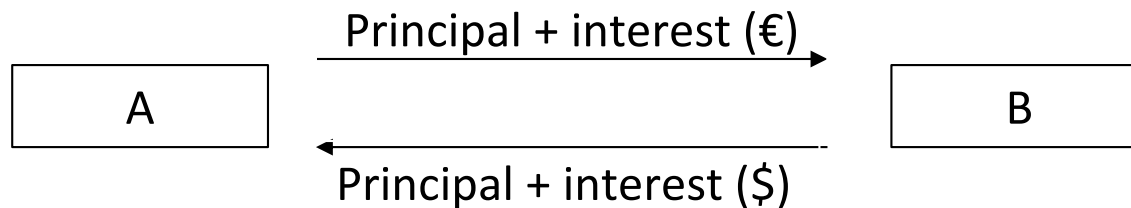
- ✓ Learn definition and terms of financial derivatives
- ✓ Get to know participants in the market for derivatives and understand reasons for their application
- ✓ Understand basic functionality of different derivative products
- ✓ Analyse the profit and loss schedules of different derivative products
- ✓ Discuss differences between different derivative products
- ✓ Apply derivative functionalities on basic examples

Swaps – Definition

- A swap is an agreement between two parties to exchange cash flows of specified financial instruments, calculated in a specific way, at specific dates in the future.
- Usually at the initiation of the contract, at least one of these series of cash flows is determined by a random or uncertain variable such as a floating interest rate, foreign exchange rate, equity price or commodity price.
- Note that only the cash flows are exchanged, not the underlying financial instruments themselves.
- A swap shall enable the contract parties to change the risk profile of the cash flows from financial assets and liabilities without actually changing the structure of assets and liabilities.

Currency Swaps – General Description

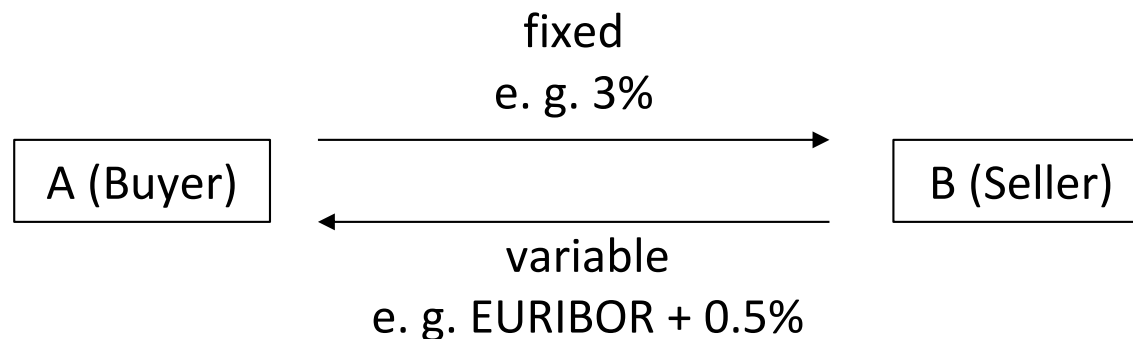
- A currency swap is an agreement in which the contract parties exchange principal and interest payments in one currency for principal and interest payments in another currency.



- The principal has to be specified in each of the two currencies and it is exchanged at the beginning and at the end of the contract.
- Currency swaps are typically used to transform a liability (or an investment) from one currency into another.

Interest Rate Swaps – General Description

- In a plain vanilla interest rate swap (IRS), the swap buyer pays fixed cash flows equal to interest on a principal for a number of years.
- In return he receives interests on the same principal for the same period at a floating interest rate.
- IRS is typically used for liabilities, but it can also swap investment cash flows.

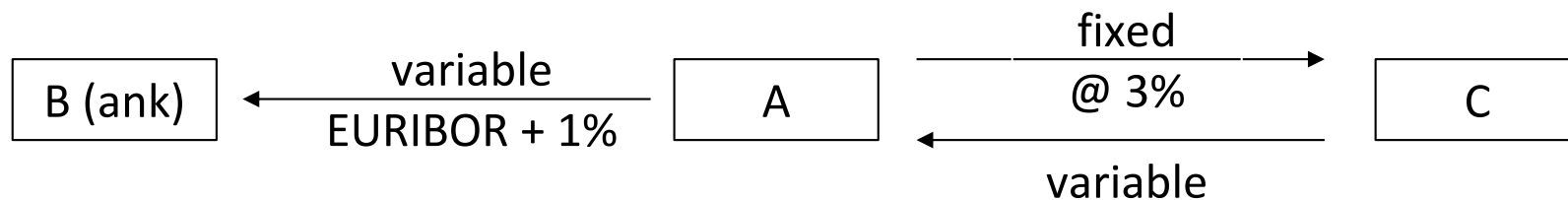


Interest Rate Swaps – General Description

- Usually two non-financial companies will not arrange a swap agreement directly. They deal with a financial institution, called market maker.
- A market maker (MM) quotes for different maturities a bid and offer a fixed-rate that he will exchange with the floating.
- There is a positive difference between these two rates, called “bid-ask spread”. It represents the market maker’s gain.
 - Bid: the fixed-rate the MM would pay (receiving the floating-rate)
 - Ask: the fixed-rate the MM would receive (paying the floating-rate)

Interest Rate Swaps – Example (1)

- A has a loan of 100 Mio. € @ EURIBOR + 1.0% margin.
- C (e. g. another Bank) offers to swap the variable interest against a fixed interest of 3%:



EURIBOR	1.5%	2.0%	2.5%	3.0%	3.5%
A => B	-2.5 Mio. €	-3.0 Mio. €	-3.5 Mio. €	-4.0 Mio. €	-4.5 Mio. €
A => C	-3.0 Mio. €	- 3.0 Mio. €	-3.0 Mio. €	-3.0 Mio. €	-3.0 Mio. €
<u>C => A</u>	<u>+1.5 Mio. €</u>	<u>+2.0 Mio. €</u>	<u>+2.5 Mio. €</u>	<u>+3.0 Mio. €</u>	<u>+3.5 Mio. €</u>
= Total A	-4.0 Mio. €	-4.0 Mio. €	-4.0 Mio. €	-4.0 Mio. €	-4.0 Mio. €

Interest Rate Swaps – Example (2)

- Some companies have a comparative (=relative) advantage when borrowing fixed-rate loans and other companies have a comparative advantage when borrowing floating-rate loans.

Company	Fixed loan rates	Floating loan rates
A	10.0 %	EURIBOR + 0.3%
B	11.2 %	EURIBOR + 1.0%

- A has a better credit rating and obtains better conditions, but it appears to have a comparative advantage in fixed-rate markets, while B has a comparative advantage in floating-rate markets.
- If A wants to borrow a floating loan and B wants to borrow a fixed loan they should swap their interest obligations.

Interest Rate Swaps – Example (2)

- There is an opportunity to deal a swap agreement:
 - A borrows at fixed-rate (10%)
 - B borrows at floating-rate (6-months EURIBOR +1.0%)
- And then they swap these cash flows:
 - A pays 10.0% to his lender, pays EURIBOR to B and receives 9.95% p. a. from B.
 - B pays EURIBOR +1.0% to his lender, 9.95% to A and receives EURIBOR from A.

